



Artificial intelligence meets epigenetics: a new frontier in precision oncology

Ruby Srivastava

To cite this article: Ruby Srivastava (2026) Artificial intelligence meets epigenetics: a new frontier in precision oncology, Epigenetics Reports, 4:1, 2614541, DOI: [10.1080/28361512.2026.2614541](https://doi.org/10.1080/28361512.2026.2614541)

To link to this article: <https://doi.org/10.1080/28361512.2026.2614541>



© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.



Published online: 14 Jan 2026.



Submit your article to this journal [↗](#)



Article views: 741



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 1 View citing articles [↗](#)

REVIEW ARTICLE



Artificial intelligence meets epigenetics: a new frontier in precision oncology

Ruby Srivastava 

Department of Chemistry, Indian Institute of Technology Bombay, Mumbai, India

ABSTRACT

Epigenetics involves heritable and stable changes in gene expression that occur without altering the DNA sequence, primarily through chemical modifications to DNA and alterations in chromatin structure. Dysregulation of these epigenetic processes is widely recognized as a hallmark of cancer, influencing tumor initiation, progression, and recurrence. Consequently, characterizing cancer-associated epigenetic alterations has become a major research focus, both to understand disease mechanisms and to identify therapeutic targets. Importantly, many epigenetic changes are reversible, making them attractive candidates for targeted therapies. However, the complexity and volume of epigenomic data; spanning DNA methylation, histone modifications, chromatin accessibility, and gene expression demanding advanced computational approaches for meaningful interpretation. Artificial intelligence (AI), particularly machine learning and deep learning, has emerged as a powerful solution for extracting biologically relevant patterns from such high-dimensional datasets. These approaches enable prediction of disease risk, classification of tumor subtypes, discovery of novel regulatory mechanisms, and development of patient-specific therapeutic strategies. Rather than simply enhancing traditional computational biology, AI fundamentally expands the analytical and predictive capabilities of epigenetic research. This review highlights current applications of AI in epigenomics-based precision oncology, along with ongoing challenges and future directions in integrating advanced computational modeling with biological insight.

ARTICLE HISTORY

Received 22 July 2025
Revised 28 December 2025
Accepted 5 January 2026

KEYWORDS

Epigenetics; cancer;
histones; machine learning;
deep learning

Introduction

From massive-scale language models to image recognition networks, artificial intelligence (AI) has grown into a powerhouse of pattern-recognition, prediction and automation across industries. In fields like autonomous driving, medical imaging and finance, AI systems now routinely sift through millions or billions of data points, learn subtle relationships in high-dimensional space, and output predictions, classifications or even generative designs at scales far beyond what manual or conventional algorithmic approaches could handle [1]. In contrast, ‘classic’ computational biology applies defined mathematical or statistical models (regressions, clustering, network analyses) to biological data effectively, but often limited when facing the volume, heterogeneity and complexity of modern omics datasets. In biomedical science, the impact of AI can be seen, as deep learning (DL) has enabled breakthroughs in protein-structure prediction, accelerating tasks that previously required years of experiment [2]. AI is also applied in drug-repurposing, image-based diagnostics and multi-omics integration. These applications illustrate how AI brings power (ability to handle massive data, learn complex non-linear relationships) and scale (processing high-throughput data, integrating diverse data types, automating routine tasks) in ways that go beyond traditional computational biology. When applied to epigenetics, the study of gene-regulation mechanisms beyond DNA sequence (such as methylation, histone modifications and chromatin conformation), AI offers distinct advantages. Epigenetic datasets are large, multi-layered and noisy; patterns often lie hidden in the interplay of methylation, chromatin state, non-coding RNA and gene expression (GE). Traditional

CONTACT Ruby Srivastava  amitruby1@gmail.com,  30006613@iit.ac.in  Department of Chemistry, Indian Institute of Technology Bombay, Mumbai, India

© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.
This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (<http://creativecommons.org/licenses/by-nc/4.0/>), which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

Epigenetics is broadly defined as the study of heritable changes in GE that occur without alterations to the underlying DNA sequence, which depends on an individual's genotype, age, and lifestyle factors such as diet, alcohol use, and smoking [4]. The term 'epigenetics' was first introduced in 1942 by Conrad H. Waddington, which is derived from the Greek word *epigenesist* [5]. Epigenetics originally describes how genetic processes influence development. In eukaryotic cells, genomic DNA is wrapped around histone proteins to form nucleosomes, the fundamental units of chromatin comprising 146 base pairs of DNA wound around an octamer of four core histones (H3, H4, H2A, and H2B). These nucleosomes further fold into higher-order chromatin structures. The spatial organisation of chromatin plays a critical role in nucleosome positioning, recruitment of transcription regulators (TRs), chromatin accessibility, and GE regulation [2]. Tightly packed chromatin typically inhibits mRNA transcription, while more open structures allow TRs and RNA polymerase to access DNA. Epigenetic mechanisms primarily govern these changes in chromatin architecture. Chromatin remodelling, driven by epigenetic modifications such as DNA methylation, histone modifications, and RNA-mediated mechanisms, controls the folding and accessibility of chromatin fibres. Post-translational modifications of histone tails, such as acetylation and methylation, serve as markers for gene activation or repression [6]. Chromatin can store and transmit these epigenetic codes, either through DNA methylation or histone modifications during cell division. Consequently, disruptions in these processes are a hallmark of nearly all cancers. Aberrant GE and irregular chromatin structure, influenced by misregulated epigenetic factors, can drive the transformation of normal cells into malignant ones [6]. Epigenetic regulation orchestrated by protein complexes is generally classified as *writers*, *readers*, and *erasers*. Writers add specific chemical modifications to DNA or histones, readers (such as methyl-CpG-binding domain proteins) recognise and interpret these modifications, and erasers remove them to dynamically regulate GE. See [Figure 1](#).

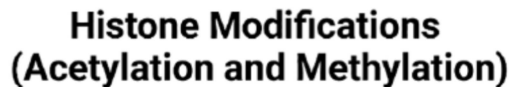


Figure 1. Representation of histone modifications (Acetylation and Methylation) with readers, writes and Erasers. (Figure Created using Biorender) [7].

Dysregulation in these mechanisms, including abnormal DNA/RNA methylation and faulty post-translational modifications, is commonly observed across cancer types. Among epigenetic mechanisms, DNA methylation plays a central role, and its process involves the addition of a methyl group to the C5 position of cytosine, forming 5-methylcytosine (5-mC), which can inhibit transcription factor binding or attract repressive proteins, thereby modulating GE. DNA methylation is regulated by DNA methyltransferases, methyl-binding proteins, and chromatin remodelers [6].

The concept of *epigenome-wide association studies* (EWAS), introduced in 2011, is the relationship between epigenetic variants and phenotypic traits [8]. Analysing methylation patterns is necessary because DNA methylation plays a crucial role in regulating GE without altering the DNA sequence itself. Aberrant methylation is associated with the onset and progression of various complex diseases including cardiovascular disorders, neurological and psychiatric conditions, and cancers [9]. Understanding these methylation changes helps identify disease mechanisms, discover biomarkers for early diagnosis, and develop targeted therapeutic strategies [10–12]. Therefore, studying methylation provides valuable insights into disease risk, progression, and response to treatment [13]. Epigenetic datasets, particularly those involving genome-wide methylation profiling, are extremely large. Whole-genome bisulfite sequencing (WGBS), for instance, can generate hundreds of gigabytes of data per sample, covering over 28 million CpG sites in the human genome. Large population studies involving thousands of samples can produce terabytes of data. In addition to methylation levels, such datasets often include metadata like GE, histone modifications, and clinical variables, further increasing their complexity and computational demands [14]. Earlier, methylation analysis was limited by high costs, low-throughput methods (such as methylation-specific PCR), and limited computational capacity to process genome-scale data. Advances in high-throughput sequencing technologies as next-generation sequencing (NGS), microarray platforms, and bioinformatics tools have drastically reduced costs and improved accuracy and speed. At the same time, advancements in computing power, storage, and cloud-based analysis pipelines now make it feasible to handle, process, and interpret massive epigenetic datasets efficiently. Together, these technological improvements have transformed methylation analysis from a niche laboratory method into a scalable, routine component of molecular disease research [15–18]. As epigenetic modifications influenced by the 3D genome architecture are reversible, reversing epimutations has become a primary therapeutic strategy. Small molecule inhibitors are being developed to target these alterations [19]. Technologies such as NGS and AI have significantly deepened our understanding of epigenetic regulation in cancer [20,21]. The most widely used method for assessing genome-wide DNA methylation is the Illumina Human Methylation Infinium BeadChip microarray [22], with other methods like pyrosequencing and methyl-sensitive restriction enzyme assays also utilised, each with unique advantages and limitations [23]. The key constraints of the Illumina methylation-array platform (e.g., the Infinium HumanMethylation450K & EPIC Bead Chips) included limited genomic coverage, probe-specific technical and biological artefacts, single-probe per region limitation, probe annotation and mapping issues and biases in assay design; which affect the machine learning (ML)/DL model performance with following manners [24,25]. A limited coverage means fewer features that capture the full methylome and less information to feed ML/DL models towards potentially lower accuracy. Technical artefacts and unreliable probes reduce feature quality and noisy inputs degrade model training and prediction. Models may learn array-specific artefacts rather than true biological signals, harming transferability to other platforms (e.g., WGBS) or datasets. Sparse or biased probe sets may limit the interpretability of model features, and annotation ambiguities reduce trust in model findings across populations/contexts [26–28]. If key CpG sites (e.g., enhancers, non-CpG methylation) are missing, ML/DL models may fail to identify important disease-linked methylation patterns [29]. Despite the constraints, the Illumina platform remains a cornerstone in epigenetic research. The clinical application of epigenetic biomarkers has seen considerable progress, especially in oncology. For instance, the *mSEPT9* biomarker is FDA-approved for colorectal cancer screening [30]. Additionally, DNA methyltransferase inhibitors (DNMTs) like azacitidine and decitabine were among the first epigenetic drugs approved by the FDA to treat myelodysplastic syndromes (MDS) [31]. The integration of ML and DL in NGS with AI technologies has accelerated the discovery of novel biomarkers [32]. These computational approaches are particularly effective at deciphering patterns within large, complex datasets, providing therapeutics for the cancer targets [33].

Artificial intelligence in epigenetics

The advancement of personalised medicine is closely related to the integration and analysis of diverse ‘omics’ data alongside individual clinical and patient information [32,34]. AI, particularly through ML, provides powerful tools for designing and training algorithms capable of learning from and acting on large datasets [22]. AI plays a vital role in organising, integrating, and interpreting complex, high-dimensional biological data [35]. **Table 1** represents a layout for AI vs ML vs DL, with definition, examples in biology with advantages and limitations.

Increasingly, AI/ML/DL approaches are proving essential for detecting polygenic diseases by uncovering intricate patterns within expanding genetic and epigenetic datasets [36]. A single human whole-genome sequencing (WGS) dataset (raw reads, alignments, variant calls) can reach ~0.5 terabytes (TB) per genome in large-scale projects. According to the National Human Genome Research Institute (NHGRI), a human genome sequence alone is ~200 gigabytes (GB) and the global genomics sequence-data generation is projected at ~40 exabytes by 2025, every ~7 months in some estimations. For scale of epigenetics (specifically DNA methylation/epigenome) data, the Infinium Methylation EPIC cover ~850,000 CpG sites per sample [37]. For example: the EWAS Data Hub, aggregates > 75,000 samples of methylation-array data across many tissues and contexts. Data file sizes for bisulfite-sequencing methylation (WGBS) may range from ~700 MB to 1 GB per sample (just for certain formats) in differential methylation analyses [38]. Integrating genomics (DNA sequence, variants, structural variation) with epigenetics enables richer models of regulatory mechanisms, disease associations, and biomarker discovery. Because a single WGS genome may require ~0.5 TB of storage, and large cohorts may include thousands of genomes, the storage footprint quickly rises into tens or hundreds of TB, or even exabyte scale globally and for methylation array data, while per-sample storage is much smaller (e.g., ~10s of MB for array output files) the aggregated data across thousands of samples becomes large. ML/DL training typically requires loading many samples and many features into memory (RAM) or into batches streamed from disk [38]. For large (genomics-epigenetics) integration, not only enough disk is used to store raw and processed data, but also fast input/output (I/O) (solid-state drives or distributed file systems), enough RAM to load features (or enable streaming/batching), and sufficient compute (CPU/GPU) for training ML/DL models are needed. Without sufficient memory/storage/I/O, ML/DL workflows may bottleneck in data loading and feature extraction; training may run out of memory, and I/O can dominate runtime rather than compute [39,40].

Machine learning (ML) in epigenetics

ML helps computers find patterns in data and make predictions based on them [40]. The rise of large-scale biological data has made ML especially powerful in research areas like genomics and epigenetics. In most ML studies, the available data are divided into three parts: a training set (used to teach the model), a test set (used to check how well it performs), and a validation set (ideally taken from a completely different source to test how well the model generalises). When a separate validation dataset is not available, researchers often use k-fold cross-validation, where the data are randomly split into k parts [41]. The model is then trained and tested multiple times using different combinations of these parts to get a more reliable measure of performance. The choice of performance metrics depends on whether the dataset is balanced or imbalanced. For balanced datasets, where all classes have a similar number of samples with common measures include accuracy, sensitivity, specificity, and precision [40,41]. However, when one class has far

Table 1. A layout for AI vs ML vs DL, with its definition, examples in biology, advantages and limitations.

Level	Definition	Example in Biology	Advantages	Limitations
AI	Broad field enabling machines to mimic human intelligence.	Predictive modelling for disease outcomes or drug responses.	Automates reasoning, integrates diverse data types.	Requires large datasets; interpretability challenges.
ML	Subset of AI that learns patterns from data without explicit programming.	Classifying cancer subtypes using methylation or transcriptomics data.	Handles large datasets, discovers complex associations.	May overfit or perform poorly on unseen data; feature engineering required.
DL	Subset of ML using multi-layered neural networks to model nonlinear relationships.	Inferring enhancer–promoter interactions or single-cell epigenetic states.	Learns directly from raw data, high predictive power.	Data-hungry, computationally intensive, limited interpretability.

more samples than the other (imbalanced data), metrics such as the F1-score, area under the receiver operating characteristic curve (AUC), and Cohen's Kappa are more informative [40–42].

The details regarding performance metrics used for the ML models is given in Table 2.

In epigenetic research, both supervised and unsupervised ML methods are widely used to analyse complex biological data [43]. Supervised learning is typically applied to classification and prediction problems, where models learn from labelled data. Common supervised algorithms include linear and logistic regression, support vector machines (SVMs), random forests (RFs), and least absolute shrinkage and selection operator (LASSO) regression [43]. These approaches have been successfully employed to classify and predict a range of diseases using epigenetic data, such as metastatic brain tumours, prostate cancer, coronary heart disease, neurodevelopmental disorders, and central nervous system tumours [44–48]. In contrast, unsupervised learning methods analyse data without predefined labels to uncover intrinsic structures or patterns [49]. While these algorithms can reveal correlations among variables, they do not directly determine biological meaning or causal relationships. Common unsupervised methods include k-means clustering, hierarchical clustering, principal component analysis (PCA), and partial least squares discriminant analysis (PLS-DA) [50,51]. These methods are often used to identify distinct DNA methylation subtypes that help classify diseases and can then be correlated with clinical or molecular characteristics. Within clustering approaches, hierarchical clustering (HC) and partition clustering (PC) group data points based on similarity measures such as Euclidean distance [52]. A more robust alternative, partitioning around medoids (PAM), minimises sensitivity to outliers but is computationally demanding, which limits its application to very large datasets. Validating cluster stability is therefore crucial to ensure reliable results. More advanced ML models, such as Gaussian Processes (GPs), Mixtures of Hierarchical GPs (MOHGP), and Multifactor Dimensionality Reduction (MDR), have also been applied to gene expression data [53]. Additionally, PCA remains a key tool for reducing data dimensionality while retaining important patterns, facilitating pattern recognition, outlier detection, and class definition, especially for high-dimensional datasets like those found in epigenomics. Overall, combining supervised and unsupervised ML strategies has proven highly effective in various diagnostic and clinical applications, including mutation detection and prediction, biomarker discovery and prognosis, disease classification and aetiology exploration, and the design of precision therapies [54,55]. However, a major limitation of unsupervised learning is that it does not inherently assign biological meaning or significance to the patterns or clusters it identifies [56]. Additionally, unsupervised algorithms are highly sensitive to data noise. The presence of irrelevant or low-quality data can lead to incorrect clustering. For this reason, rigorous data pre-processing is essential to ensure accuracy and reliability of analysis outcomes [57].

Table 2. List of various performance measures used for machine learning models.

Metric	Definition	Description/Interpretation
True Positive (TP)	Correctly predicted positive cases	Number of samples correctly identified as positive (e.g., disease present).
True Negative (TN)	Correctly predicted negative cases	Number of samples correctly identified as negative (e.g., disease absent).
False Positive (FP)	Incorrectly predicted positive cases	Number of samples incorrectly labelled as positive.
False Negative (FN)	Incorrectly predicted negative cases	Number of samples incorrectly labelled as negative.
Accuracy	Proportion of correct predictions among total samples	Measures overall correctness of a model's predictions. $\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$
Precision (Positive Predictive Value)	Proportion of true positives among predicted positives	Indicates reliability of positive predictions. $\text{Precision} = \frac{TP}{TP + FP}$
Recall (Sensitivity/True Positive Rate (TPR))	Proportion of true positives correctly identified	Reflects ability to detect positive cases. $\text{Recall} = \frac{TP}{TP + FN}$
Specificity (True Negative Rate)	Proportion of true negatives correctly identified	Reflects ability to identify negative cases. $\text{Specificity} = \frac{TN}{TN + FP}$
F1-Score	Harmonic mean of precision and recall	Balances precision and recall into a single measure. $F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$
ROC-AUC (Area Under the Receiver Operating Characteristic Curve)	Area under the ROC curve plotting TPR vs. FPR (False Negative Rate)	Measures model's ability to discriminate between classes. $C = \frac{\text{Number of concordant pairs}}{\text{Total comparable pairs}}$
C-index (Concordance Index)	Fraction of all pairs of subjects whose predicted and actual outcomes are concordant	Commonly used in survival analysis; values range from 0.5 (random) to 1.0 (perfect prediction).

Epigenetic data, particularly DNA methylation, applies ML applications as DNA methylation is generally both chemically and biologically stable over time, and access to expansive data repositories such as The Cancer Genome Atlas (TCGA) [58], ENCODE [59], and the BLUEPRINT consortium [60] allows implementation of large-scale analyses of biologically meaningful, differentially methylated regions [61–63]. DNA epigenetic modifications vary in their stability depending on their biochemical nature and genomic context. 5-mC, the most studied DNA methylation mark, is generally stable and heritable through cell division, especially when located in CpG islands of gene promoters or repetitive regions, where maintenance methyltransferases (e.g., DNMT1) copy the methylation pattern during replication. In contrast, 5-hydroxymethylcytosine (5-hmC) and other oxidised derivatives produced by ten-eleven translocation (TET) enzymes are more dynamic and reversible, often representing intermediate states in active demethylation pathways [64]. Similarly, methylation at enhancer regions or non-CpG sites (CpA, CpT, CpC) tends to be more variable and cell-type specific, responding rapidly to developmental cues or environmental stimuli. Thus, promoter and repeat methylation marks are typically stable, while enhancer, gene-body, and non-CpG methylation marks are more in flux, reflecting the adaptive and context-dependent nature of the epigenome [12].

The Sequence Read Archive (SRA) also support tools such as Magic-BLAST, to facilitates the alignment of sequencing reads to reference genomes using existing sequence databases [65]. Magic-BLAST is a versatile aligner developed by National Centre for Biotechnology Information (NCBI) for mapping both RNA and DNA sequencing reads to reference genomes or transcriptomes. Magic-BLAST is optimised for RNA-seq alignment, efficiently handling spliced reads that span exon–exon junctions, making it particularly useful for transcript quantification and variant discovery in RNA data. Magic-BLAST integrates seamlessly with NCBI’s BLAST infrastructure and supports large datasets while maintaining high accuracy and low memory usage [65,66]. For DNA read mapping, widely used tools include BWA (Burrows–Wheeler Aligner) [67], Bowtie2 [68], and Hisat2 [69]. These algorithms align short or long sequencing reads to a reference genome to detect sequence variants, structural rearrangements, and coverage profiles. Accurate read mapping is crucial because the identification of genetic and epigenetic variants such as single-nucleotide polymorphisms (SNPs), insertions/deletions, and methylation marks depends heavily on the quality and precision of alignments. This is especially critical in bisulfite sequencing data, where chemical conversion of unmethylated cytosines introduces additional complexity. Specialised bisulfite aligners, such as Bismark [70], are designed to handle these modifications and ensure accurate methylation calling. Poorly mapped reads can lead to false variant calls or misinterpretation of methylation levels, underscoring the importance of robust mapping algorithms in epigenomic analysis. The large, publicly available repositories serve as valuable sources for both training and validating ML algorithms, as ML models depend on vast quantities of high-quality data for accurate predictions. It enhances the external validity and enabling assessment of clinical applicability. The integration of AI–ML with epigenomics has enabled significant progress in the diagnosis and classification of rare diseases (RDs), including various cancers, cerebral palsy, and neurodevelopmental disorders [7,71,72]. Several RDs have been associated with genetic mutations in genes regulating DNA methylation or histone modifications, such as those observed in Rett syndrome [73], and Cornelia de Lange syndrome [74]. Additionally, disruptions in the genomic imprinting process which depends on proper DNA methylation and histone modification are implicated in syndromes like Prader–Willi [75], and Beckwith–Wiedemann [76].

Long-read sequencing platforms such as PacBio and Oxford Nanopore enable the detection of long-range epigenetic interactions by sequencing extended DNA or RNA fragments in a single read. Unlike short-read technologies, these methods can directly identify base modifications without the need for amplification, offering a more precise representation of native epigenetic patterns [77]. Nanopore sequencing detects these modifications by measuring changes in electrical current as nucleic acids pass through a nanopore, whereas PacBio’s Single-molecule real-time (SMRT) sequencing monitors nucleotide incorporation kinetics to infer epigenetic states [78]. The long-read sequencing (Nanopore and PacBio) and Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR)-based epigenome editing systems (dCas9, dCas9-PPAD, and dCas9-Tet1)[79–81] create functionally annotated datasets that enable AI-driven modelling of regulatory networks and prediction of epigenetic intervention outcomes. Collectively, these innovations underscore how AI, combined with single-cell and spatial multi-omics, is transforming epigenetics from descriptive profiling to predictive, mechanistic, and therapeutic discovery, offering new insights into gene regulation, tumour heterogeneity, and personalised medicine [7,71–81].

Deep learning (DL) in epigenetics

DL has emerged as one of the most effective strategies for analysing NGS genomic data [32]. However, the performance and suitability of DL algorithms are highly dependent on the specific task and the nature of the input data. At the core of DL lies the artificial neural network (ANN), which is modelled after the human brain. Information is processed through layers of interconnected computational nodes, or ‘neurons,’ with each connection assigned a weight. In DL, the output of ANN is generated by propagating information forward through multiple layers of interconnected neurons. Each neuron receives inputs from the previous layer, applies a weighted sum, and passes the result through a nonlinear activation function (e.g., ReLU, sigmoid, or tanh). The final layer produces the network’s output, which may represent probabilities, class labels, or continuous values, depending on the task. During training, backpropagation is used to adjust the weights by minimising a loss function, thereby improving the accuracy of the predicted outputs over successive iterations [34]. Another widely used DL architecture is the Convolutional Neural Network (CNN), which is known for its superior performance in computer vision and pattern recognition tasks [82]. Autoencoders (AEs) represent another class of neural networks used for unsupervised learning [83]. Capsule Neural Networks (CapsNets) preserve spatial hierarchies by using ‘capsules,’ which are groups of neurons that represent specific entities or object parts and their properties, allowing better generalisation to novel viewpoints. Recurrent Neural Networks (RNNs) are suited for sequential data, such as time series, speech, text, and DNA sequences, due to their internal memory mechanisms. RNNs can retain relevant information from previous inputs, enabling context-aware predictions [84]. The Long Short-Term Memory (LSTM) networks [85] incorporate memory cells and gating mechanisms (input, forget, and output gates) that regulate information flow, allowing the model to capture both long- and short-term dependencies. Gated Recurrent Units (GRUs) are comparatively computationally efficient variant of LSTMs. Bidirectional GRUs (BiGRUs) further enhance performance by combining two GRUs running in opposite directions (forward and backward), allowing information from both the past and future to inform the current prediction [86]. Transformer-based models are another natural language processing model applied to biological data. These models rely on attention mechanisms rather than recurrence, comprising encoder-decoder architecture. Each encoder layer includes multi-head self-attention and feedforward sublayers, supported by skip connections and layer normalisation to facilitate stable training [87].

A notable application of DL in cancer research is MetaCancer [88], a DL framework for classifying primary and metastatic cancers. The study utilised RNA-Seq, microRNA-Seq, and DNA methylation data across 11 cancer types from TCGA. During pre-processing, CpG islands located 1500 base pairs upstream of transcription start sites were identified and aggregated using the ELMER (Enhancer Linking by Methylation/Expression Relationship) R package. Missing expression values were imputed across the datasets. MetaCancer demonstrated robust performance, achieving an AUC of 85%, an average accuracy of 90.69%, sensitivity of 73.97%, specificity of 97.63%, and precision of 93.38% [88]. The list of other pipelines/software link, DL architecture, disease(s) and omics data used to classify cancer epigenetics is given in Table 3.

Applications in DNA methylation-based cancer classification

In a unimodal study, two DL models were implemented for distinguishing malignant from non-malignant samples, and a multi-class classification model for predicting pan-cancer types based on malignancy information and shared genomic features [34]. Both models achieved consistently high performance, with the binary and multi-class classifiers attaining an impressive AUC of 90.92% and an overall accuracy of 98.33%, an F1 score of 98.91%, precision of 98.77%, and recall of 98.32%. Additionally, the binary model achieved a Matthews correlation coefficient (MCC) of 93.5% [34]. Another study proposed a ML pipeline for breast cancer prediction and identification of key methylation-driven genes using TCGA-BRCA data. Feature engineering was applied to reduce dimensionality and improve deep learning scalability. Comparative analysis of Illumina 27 K and 450 K datasets demonstrated that integrating deep learning with feature selection enhances prediction accuracy, achieving 98.75% accuracy within 13 s using 450 K markers. Enrichment analysis revealed significant biological processes and molecular functions in both datasets, with seven shared oncogene/tumour suppressor genes identified across platforms [89].

Table 3. List of Pipelines/software links, DL architecture, disease(s) and omics data used to classify cancer epigenetics.

Pipeline name/Software link	DL Architecture	Disease(s)	Omics data
Multi-omics datasets metacancer https://github.com/somayahabaradei/MetaCancers https://drive.google.com/open?id=1LsYe3ypiweox2OSmnD5LdaYMJePeozDd	Conditional Variational Autoencoder (CVAE), FClayer	11 types of cancer from The Cancer Genome Atlas (TCGA) projects	RNA-Seq microRNA-Seq, DNA methylation
CapsNetMMD https://github.com/ustcp/CapsNetMMD	Feedforward Neural Network (FNNs)	Uterine cervical cancers	DNA methylation, GE
Ddae-MLP https://github.com/vd4mmind/multiOmicsIntegrations	CapsNets	Breast Cancers	mRNA expression, DNA methylation, Two forms of DNA copy number alterations (CNAs)
DFNForest https://github.com/tuinaiao316/datasets	DAE multilayer perceptrons (MLPs)	LIHC	DNA methylation, RNA-seq, CNV
	Stacked AEs, Deep flexible neural forest (DFN Forest)	Breastinvasive carcinoma (BRICA) Glioblastoma multiforme (GBM) Ovarian cancers (OV)	GE, DNA methylation, miRNA expression
Gcbmi		Gastric cancers	
Unimodal data (DNA methylation)	DNNs	Gastric cancers	DNA methylation, GE
https://github.com/rauhlgomes19/Deep_Learning_Methylation	Sequential DNNs	Breast cancers	DNA methylation
MethylCapsNet and MethylSPWNet https://github.com/Christensen-Lab-Dartmouth/MethylCapsNet	CapsNet, MLPs, GLRM in DNNs	Central Nervous System (CNS) tumour	DNA methylation
MethylNet https://github.com/Christensen-Lab-Dartmouth/MethylNet	Variational Autoencoder (VAEs)	32 different cancers subtypes, Breast cancer	Six public DNA methylation data sets for a range of various tasks
DisMir https://github.com/XWangLabTHU/DisMir	Cable News Network (CNNs), Bidirectional Long Short-Term Memory Networks (BLSTMs)	Hepatocellular Carcinoma (HCC)	cell-free DNA methylation data
iCancer-Pred https://github.com/Huerhu/iCancer-Pred	Fully Connected (FC) layers	7 different cancers subtypes from TCGA	DNA methylation
Cancer origin predictions https://github.com/thunder001/Cancer_origin_predictions	MLP	18 different cancers origins from TCGA	DNA methylation, clinical patient informations
Multi-omics datasets https://github.com/gevaertlab/BetaVAE-Imputations	VAEs	RNA-seq, DNA methylation	RNA-seq, DNA methylation

Malformations of cortical development (MCD) are heterogeneous epilepsy-associated brain lesions with challenging diagnosis. This multi-centre study analysed genome-wide DNA methylation of 308 brain samples using EPIC 850 K arrays, including 239 FFPE MCD cases spanning 12 subtypes and non-MCD controls. Pairwise statistics, machine learning, and deep learning methods were applied for classification. A shallow neural network achieved the highest accuracy, correctly classifying all 43 independent test samples, supporting methylation-based MCD diagnostics [90]. In the study of genome-wide DNA methylation in 96 brain metastases specimens (including 30 from breast cancer), the integration of ~1,860 publicly-available methylation profiles from primary and normal tissues, unsupervised hierarchical clustering of the most variably methylated CpG sites were used and it was found that the metastatic brain tumour samples clustered by tissue of origin (breast vs lung vs melanoma) and by clinically relevant breast-cancer brain metastasis (BCBM) therapeutic subtype (hormone-receptor positive/HER2-negative, HER2-positive, triple-negative) in 'Class C' of their three-step classifier. Although the classifier in this study is ultimately supervised, the unsupervised clustering component (hierarchical clustering on methylation β -values) revealed distinct methylation sub-groups among brain metastases that corresponded to breast-cancer subtypes (and metastasis vs primary/normal tissue) i.e., methylation patterns distinguished between different disease states without prior labelling [91].

MethylCapsNet and MethylSPWNet [92] are DL-based frameworks tailored to methylation data for predicting categorical outcomes in central nervous system (CNS) diseases. MethylCapsNet uses capsule networks to group CpG sites into biologically meaningful capsules, incorporating information such as gene promoter context and CpG island annotations. MethylCapsNet uses separate MLPs for each CpG grouping (capsule) to create context-specific data representations, or embeddings. It then uses a dynamic routing process to force information from 'child capsules' (the CpG groupings) into 'parent capsules' that represent categorical outcomes, like different tumour types. MethylSPWNet is a specialised neural network architecture for analysing larger numbers of CpG contexts compared to MethylCapsNet, and routes

methylation values from CpGs into a single node that represents the entire context (e.g., a gene), creating a gene-level summary. Further it assigns a weight to each CpG based on its importance to see which specific sites within a capsule are most significant for a prediction [92]. MLPs are applied with dynamic routing algorithms to manage the interaction between child and parent capsules for accurate prediction. MethylSPWNet employs a group LASSO regression framework to retain relevant genes while filtering out noise. Both methods demonstrated outstanding accuracy (98%) and consistent precision, recall, and F1 scores (97%). Methylnet, a pre-trained variational autoencoder (VAE) model [93], was applied to extract biologically informative features for multiple tasks, including multi-output regression (e.g., age prediction and cell-type deconvolution) and classification (e.g., pan-cancer subtypes and smoking status). Methylnet is a deep learning framework for DNA methylation analysis that enables embedding construction, prediction, data generation, and discovery of hidden heterogeneity with minimal user input. Methylnet effectively captures cellular variations, identifies higher-order features distinguishing cancer subtypes, estimates biological age, and detects smoking-associated methylation patterns consistent with established findings. For age estimation, Methylnet model attained an R^2 of 0.96 and a mean absolute error (MAE) of 3.0 [93]. DISMIR was introduced to identify tumour-derived reads in cell-free DNA (cfDNA) WGBS data, estimating their proportion in plasma samples. Applied to hepatocellular carcinoma (HCC) samples, DISMIR achieved an AUC of 99.69%, recall of 93.94%, and perfect specificity (100%). The correlation between tumour size and tumour-derived read ratio was 0.88 ($P = 6.68 \times 10^{-5}$) [94]. iCancer-pred [95] was developed as a diagnostic platform to detect multiple cancer types using DNA methylation data to distinguish healthy from cancerous samples, and another for multi-class classification (seven cancer subtypes). In the ‘TCGA Hub’, DNA methylation (HumanMethylation450) data can be found and breast invasive carcinoma, colon adenocarcinoma, kidney renal clear cell carcinoma, kidney renal papillary cell carcinoma, liver hepatocellular carcinoma, lung adenocarcinoma and lung squamous cell carcinoma are chosen. The model achieved outstanding performance, with an AUC of 99.68%, accuracy of 98.37%, MCC of 92.95%, recall of 98.15%, and specificity of 99.53%. Another DL MODEL, based on VAEs, was applied to analyse DNA methylation data from lung adenocarcinoma (LUAD) and lung squamous cell carcinoma (LUSC) obtained from the TCGA [96]. Applying the advanced multi-view graph-level representation learning (MGRL) DL-explainable artificial intelligence (XAI) framework to DNA methylation data from sorted monocytes uncovers an epigenetically deregulated inflammatory response pathway, whose activity progressively increases with age. In DL-XAI model, the input data type is highly variable and depends on the application, while the absence of standardised performance metrics is also challenging. Evaluation metrics are often context-dependent, relying on a combination of quantitative measures, user studies, and domain-specific criteria [97].

DL applications in DNA methylation analysis and cancer prediction

Cancer of unknown primary (CUP) comprises rare malignancies with poor prognosis and unidentifiable tissue-of-origin. Using TCGA genome-wide DNA methylation data and ML, a 200-CpG methylation feature classifier (MFCUP) was developed to predict CUP origin. MFCUP achieved 97.2% accuracy in a validation cohort of 5923 samples across 25 cancer types. When tested on Infinium 450 K and EPIC arrays, accuracies were 93.4% and 84.8%, respectively. A targeted bisulfite sequencing panel validated on 78 FFPE samples correctly identified tissue origin in 88.5%, highlighting the diagnostic potential of specific CpG methylation biomarkers [98]. A VAE-based approach was utilised to impute missing values in GE and DNA methylation (DM) datasets within multi-omics data. The study evaluated performance under two scenarios: Missing Completely at Random (MCAR) [99] and Missing Not at Random (MNAR), using real-world inspired simulations. DeepCpG [100] was developed to predict binary methylation states of CpG sites at the single-cell level on single-cell bisulfite sequencing (scBS-seq) and single-cell reduced-representation bisulfite sequencing (scRRBS-seq) datasets from five cell types (Serum, 2i, HCC, HepG2, and mESC), achieving AUC values ranging from 82% to 97%, with an average AUC of 92.4%. BiLSTM-5Mc [101] is used to detect 5 mC modifications in nucleotide sequences on small-cell lung cancer (SCLC) data, with an AUC of 96.35%, accuracy of 93.03%, sensitivity of 86.61%, specificity of 93.74%, and a Matthews correlation coefficient (MCC) of 63.84%.

CpG Transformer [102] is designed to handle incomplete methylation matrices with inputs including methylation data, genomic positions, and surrounding DNA sequences efficiently captures autocorrelations

between adjacent CpG sites, enhancing prediction robustness. D-GPM [103], a DL model for large-scale multitask learning, utilised promoter methylation data from the Illumina HumanMethylation450K array in TCGA, incorporating data from 902 landmark genes and 21,645 target genes with a Mean Absolute Error (MAE) of 0.03 and Pearson Correlation Coefficient (PCC) of 0.82. MRCNN [104], a model designed for genome-wide prediction of CpG site methylation levels used WGBS data, utilised 400 bp DNA sequence fragments centred at target CpG sites to learn associations between sequence features and methylation ratios, expressed as continuous values between 0 and 1. The MRCNN model reported average regression errors for hypermethylation, hypomethylation, and intermediate methylation statuses with Root Mean Square Error (RMSE) and MAE values of 0.21 and 0.19, respectively [104].

Advanced DL models for DNA methylation analysis, cancer diagnosis, and prognosis

A recent INTERACT model [105], is developed to predict the effects of genetic variation on DNA methylation in the human brain. Performance was assessed across four tissue types; brain, blood, buccal, and saliva; achieving a Spearman correlation of 0.78, RMSE of 0.15, and an AUC of 82.5%. Various models as ML_ordCOX [106] Stacked Denoising Autoencoder (SdA) [107] were designed for survival analysis and subtypes classification. In lung adenocarcinoma prognosis, a DL-based model incorporated multi-omics data into a covariate Cox proportional hazards (Cox-PH) framework, employing LASSO regression for feature selection by minimising log partial likelihood. Validation on an external dataset showed promising prognostic capacity, with a log-rank test p -value of 9.3×10^{-3} and a concordance index (C-index) of 0.64 [108]. The MFAE model applied DL techniques to extract meaningful features and integrate multi-omics data with biological networks, achieving average AUC values of 78% and average precision of 70% for both bladder urothelial carcinoma and brain lower-grade glioma [109]. The i-Modern method [110] proposed an integrated matrix to represent high-dimensional multi-omics profiles per patient, enabling the discovery of key multi-omics signatures contributing to glioma stratification, with an exceptional result, with AUC of 99%, accuracy of 98%, sensitivity of 99%, and specificity of 96% using a fully connected (FC) layer-based prediction architecture. An autoencoder (AE) model was employed to integrate DNA methylation, RNA-seq, and miRNA-seq data from colorectal cancer (CRC) patients. This approach involved concatenating the three data matrices and reducing their dimensionality to produce a lower-dimensional representation indicating their potential utility in CRC prognosis. Evaluation of the model yielded a concordance index (C-index) of 0.78 and a log-rank test p -value of 1.53×10^{-7} [111].

In another study, AE were utilised to explore the relationship between non-small cell lung cancer (NSCLC) subtypes specifically LUAD and LUSC, and survival outcomes [112]. Logistic regression results revealed a broad range of AUC scores (43% to 99%) across different omics data matrices. Notably, k-means clustering with $K = 10$ produced a significant survival difference (log-rank test: $P = 0.003$) [113]. Another CapsNetMMD, a deep learning framework using capsule networks was used to integrate multi-omics data including mRNA expression, z-scores, DNA methylation, and copy-number alterations—to transform breast cancer gene identification into a supervised classification task. This approach outperforms existing machine learning methods, identifying prognostic genes with potential biomarker value for breast cancer research [114]. For unimodal DNA methylation data, a deep survival EWAS method was developed to uncover blood-based methylation patterns associated with time-to-diagnosis (TTD) in breast cancer to model the relationship between aggregated methylation features and TTD [115]. The CapsNetMMD method has been designed for multi-omics analysis in breast cancer by integrating mRNA expression, DNA methylation, and two types of DNA copy number alterations (CNAs) [116]. The CapsNetMMD approach builds gene feature matrices and incorporates known breast cancer genes from resources such as DisGeNET for supervised classification using integrated multi-omics data, achieving an AUC of ~94.6%, sensitivity of ~75.8%, and specificity of ~95% in identifying breast cancer-related genes [117]. A DL model termed DDAE-MLP was proposed for predicting GE values and for classifying tumour versus normal samples in liver hepatocellular carcinoma (LIHC) with an excellent performance, with an accuracy of 95.1%, precision of 96%, recall of 95%, and an F1 score of 95% on the classification task [118]. Another study proposed the HI-DFNForest model, which integrates gene expression, miRNA expression, and DNA methylation to classify cancer subtypes in TCGA BRCA, GBM, and OV datasets [119]. Other studies included AltumAge, a DL-based neural network developed using 142 publicly available human tissue

datasets, which enhances the accuracy and interpretability of epigenetic clocks beyond traditional linear regression models [120].

Recent advances in epigenetic profiling technologies, particularly single-cell and spatial multi-omics, have created unprecedented opportunities for applying AI to decode cellular heterogeneity and regulatory complexity in tumours [121]. Techniques such as single-cell ATAC-seq (scATAC-seq) [122] and its enhanced versions single-cell combinatorial indexing (sci-ATAC-seq) and TenX Combinatorial Indexing ATAC-seq (txci-ATAC-seq) [123] enable large-scale mapping of chromatin accessibility at single-cell resolution, providing high-dimensional datasets ideal for AI-driven pattern recognition and feature extraction. Similarly, integrative approaches like Sango, which combine genomic sequence data with scATAC-seq, already employ ML frameworks to overcome challenges of data sparsity and dimensionality reduction in single-cell datasets [124]. In single-cell DNA methylation profiling, methods such as single-cell bisulfite sequencing (scBS-seq), single-nucleus methyl Cytosine sequencing 2 (snmC-seq2) [125], generate detailed methylome maps that can be analysed using DL architectures to uncover hidden relationships between methylation signatures and disease phenotypes [126]. Furthermore, single-cell multi-omics platforms such as scNMT-seq (simultaneously assessing nucleosome positioning, DNA methylation, and transcription) and frameworks like MOFA+ [127], 'MINDS', and Directional Pathway Mapping (DPM) integrate multi-layered data using AI and ML-based models to identify latent biological factors and regulatory pathways [128,129]. Online tools such as Multi-omics analysis sandbox toolkit [130], playOmics [131], and MultiOmicsAgent (MOAgent) [132] also apply ML algorithms to discover biomarkers, predict patient outcomes, and model tumour heterogeneity. AI has been particularly impactful in multimodal data integration, where CNNs and transformer-based architectures generate unified cellular representations from single-cell RNA sequencing (scRNA-seq) and scATAC-seq data, improving resolution, clustering accuracy, and interpretability [133,134]. These models are invaluable for linking epigenetic modifications to GE and predicting functional consequences of chromatin state changes across cell populations. In addition, proteogenomic and metabolomic integration provides new frontiers for AI applications. By combining genomic, transcriptomic, and proteomic data, AI frameworks can identify epigenetically regulated protein targets critical to cancer progression, while mass spectrometry-based proteogenomics supports the detection of HLA-bound peptides, informing cancer immunotherapy strategies. Likewise, AI-enabled multi-omics integration of metabolomic and microbiome data has revealed causal links between metabolic pathways, microbial metabolites, and epigenetic regulation, highlighting potential therapeutic targets for personalised interventions [133].

Emerging spatial omics platforms such as Slide-seq, Slide-seqV2, and Spatial Epigenome Sequencing further enhance the role of AI by generating spatially resolved epigenomic maps that require sophisticated AI algorithms for three-dimensional tissue modelling and spatial transcriptome–epigenome correlation [129]. These technologies capture the localisation of histone modifications (e.g., H3K27me3, H3K4me3, H3K27ac) within tissue architecture, providing high-content data for AI-assisted spatial pattern recognition and disease state classification [125].

Advanced computational methods for epigenetics

The growing complexity of epigenomic data in cancer research has catalysed the development of advanced computational techniques to extract meaningful biological insights from large-scale, multidimensional datasets. DeepCpG employs a hybrid CNN-RNN model to predict single-cell DNA methylation, uncovering cell type-specific epigenetic signatures relevant to cancer [100]. To enhance interpretability, newer methods incorporate attention mechanisms and layer-wise relevance propagation to better understand the biological features influencing model predictions. The graph-based algorithms and tensor factorisation frameworks are being applied to integrate diverse epigenomic data types, for the discovery of pan-cancer epigenetic signatures. Multi-omics Imaging Integration Toolset (MIIT) tools facilitate spatial integration of epigenomic data, enriching analyses of tumour heterogeneity and gene regulation mechanisms [66]. Bayesian models are also employed to infer causal relationships between epigenetic modifications and GE, accounting for uncertainty inherent in biological datasets [129]. Time-series analyses and trans-dimensional Markov Chain Monte Carlo methods offer insights into dynamic epigenetic changes during cancer progression and treatment response [133]. Other strategies include network-based models and

specialised algorithms such as Spider (Signalling Protein Interaction Dynamic Extractor and Regulator), which use multiscale network modelling to contextualise regulatory changes in cancer [134].

The cloud-based platforms, including Google Cloud, Amazon Web Services, and the Seven Bridges Cancer Genomics Cloud, offer scalable, secure environments and access to extensive bioinformatics toolsets [135]. Despite these advances, challenges remain regarding data quality, integration, and ethical concerns, emphasising the need for robust validation frameworks and interdisciplinary collaboration. Notable methods for single-cell and spatial epigenomic analysis include: ScRRBS [136], applied in chronic lymphocytic leukaemia, ScBS-seq, used for profiling circulating tumour cells [137], and ChIP-seq, tailored for breast cancer research [137–139].

Bioinformatics tools

Various bioinformatics tools are available for prediction of all kinds of mechanisms; for analysing methylation data [140]. Visualisation of DNA methylation data, especially when integrated with GE profiles, poses unique challenges. Techniques such as heatmaps, PCA plots, beta-value histograms, and sample-to-sample density scatter plots are commonly used [141]. However, methylation analysis at the gene level is complicated by the variability in CpG probe locations, especially across genes with multiple transcripts. Choosing a single probe (e.g., based on genomic region such as TSS200 or 5'UTR) is often suboptimal due to noise, missing data, and variability in methylation patterns across probes [142]. A better approach is to visualise all CpG probes associated with a gene using boxplots across sample groups (e.g., tumour vs. normal), which provides a more comprehensive view of methylation variability. These visualisations can be generated using R, MATLAB, Python, and Julia, and most of the pipelines listed in Table 4 are supported by these tools [143,144]. Online platforms also offer powerful visualisation tools as DNMIVD (DNA Methylation Interactive Visualisation Database) [145], Smart App [146], Wanderer (CpG probe-level TCGA methylation visualisation) [147], MEXPRESS (clinical annotation overlays) [148] and MethSurv (survival analysis based on methylation for 25 cancer types) [149]. The DNA methylation tools include MethDB [150], meth PrimerDB [151], MethTools [152], MeInfoText [153], MethyCancer [154], DiseaseMeth [155], MethylomeDB [156], MethBase [157], NGSmethDB [158], DevMouse [159], PCMDB [160], Ddmgd [161], MethHC [162], MethBank [163], Msuite2 [164], DMRichR [165], CpGMe

Table 4. Various pipelines and bioinformatics tools used in epigenetics studies (All Types of Mechanisms, DNA Methylation and Post-translational histone modifications).

S.N.	Mechanism	Bioinformatics tools
1.	All Types of Mechanisms	1. NCBI Epigenomics(www.ncbi.nlm.nih.gov/epigenomics), 2. ENCODE(https://www.encodeproject.org/), 3. PEpiD(http://wukong.tongji.edu.cn/pepid), 4. Metalprint(http://bioinfo.hrbmu.edu.cn/Metalprint), 5. EpiFactors(http://epifactors.au-tosome.ru), 6. GED(http://gametsepi.nwsuafilmz.com) 7. TCGA (https://portal.gdc.cancer.gov/) 8. DNA methylation database (https://nanome.jax.org)
2.	DNA Methylation	1. MethDB (http://www.methdb.de), 2. meth PrimerDB(https://www.hs-ls.pitt.edu/obrc/), 3. PubMeth(http://www.pubmeth.org), 4. MeInfoText(http://mit.lifescience.ntu.edu.tw/), 5. MethyCancer (http://methycancer.psych.ac.cn), 6. DiseaseMeth(http://bio-bigdata.hrbmu.edu.cn/diseasemeth/), 7. MethylomeDB(http://epigenomics.columbia.edu/methylomedb/index.html), 8. MENT(http://mgcr.kribb.re.kr:8080/MENT/), 9. MethBas(http://smithlab.usc.edu/methbase/), 9. NGSmethDB (http://bioinfo2.ugr.es/NGSmethDB), 10. DevMouse(http://db.edbc.org/DevMouse/), 11. PCMDB(http://crdd.osdd.net/ra_g_hava/pcmdb/), 12. Ddmgd(http://www.cbrc.kaust.edu.sa/ddmgd/), 13. MethHC(http://MethHC.mbc.ntu.edu), 14. MethBank(http://bigd.big.ac.cn/methbank/) 15. MethGET (https://paoyang.ipmb.sinica.edu.tw/Software.html). 16. Msuite2 (https://github.com/hellosunking/Msuite2/) 17. DMRichR (https://github.com/ben-laufer/DMRichR) 18. CpG_Me(https://github.com/ben-laufer/CpG_Me)
3.	Post-translational histone modifications	1. Histone Database(http://research.nhgri.nih.gov/histones/), 2. ChromatinDB(http://www.bioinformatics2.wsu.edu/ChromatinDB), 3. Hhmd(http://bioinfo.hrbmu.edu.cn/hhmd), 4. Hlstone (http://www.iiserpune.ac.in/coee/histome/), 5. dbHiMo(http://hme.riceblast.snu.ac.kr/)

[166]. The post-translational histone modification tools include Histone Database [167], ChromatinDB [168], Hhmd [169], PubMeth [170] and dbHiMo [171].

Challenges in advanced epigenetic studies

Despite significant progress, several challenges persist. The successful application of machine learning (ML) in biological research depends heavily on access to large, high-quality datasets that enable the development of robust and generalisable models. However, one of the major challenges lies in the interpretability of deep learning algorithms, which often function as ‘black boxes,’ limiting our understanding of the biological rationale behind their predictions. Moreover, processing vast and complex datasets demands significant computational power, making the implementation of such models resource-intensive [172]. Another persistent issue is data heterogeneity and noise, as integrating multiple data types such as genomic, transcriptomic, and epigenomic information while handling missing values and variability remains a considerable hurdle. Batch effects, arising from differences in experimental conditions or protocols, can further obscure true biological signals and lead to misleading conclusions. Finally, the lack of standardised ground-truth datasets for benchmarking hinders fair comparison, validation, and reproducibility of ML models across studies, emphasising the need for well-curated and universally accepted reference datasets in the field [173,174].

Future prospects in cancer epigenetics

Epigenetics plays a fundamental role in tumour biology, with mechanisms such as DNA methylation, histone modification, chromatin remodelling, and noncoding RNAs serving as central regulators of gene expression and cellular behaviour. Recent technological advancements, including single-cell epigenomics and CRISPR-based epigenetic editing, are revolutionising cancer research by providing unprecedented precision in mapping and manipulating epigenetic landscapes [175]. Therapeutically, epigenetic drugs (epi-drugs) have demonstrated significant potential, particularly in haematologic malignancies, though enhancing their efficacy in solid tumours remains a major challenge. Combining epi-drugs with immunotherapies or conventional chemotherapies has emerged as a promising strategy to improve clinical outcomes and overcome resistance mechanisms. In drug development, there is an increasing demand for more potent and selective epigenetic inhibitors that minimise off-target effects and toxicity. Personalised medicine approaches, which tailor treatments based on individual epigenomic profiles, further hold great promise for optimising therapeutic efficacy. However, given the complex interplay between genetic and epigenetic alterations, as well as the influence of the tumour microenvironment, a deeper understanding of these interactions is essential. Finally, ethical considerations regarding the potential for heritable changes induced by epigenetic therapies must be carefully evaluated to ensure the responsible advancement of this rapidly evolving field [176,177]. The future of AI in epigenetics is poised to redefine how we decode the regulatory complexity of the genome. Recent strides such as deep learning-based prediction of DNA methylation landscapes, integration of multi-omics data for chromatin state inference, and transformer models capable of learning context-specific epigenetic signatures demonstrate AI’s potential to move beyond descriptive analyses toward predictive and mechanistic understanding [178]. However, realising this promise requires addressing key challenges: the need for standardised, high-quality multi-omics datasets; improved model interpretability to ensure biological validity; and greater integration of temporal and spatial epigenomic data to capture dynamic regulatory events. Future directions should emphasise explainable AI, cross-species transfer learning, and the incorporation of single-cell and long-read sequencing data to achieve a holistic view of epigenetic regulation. As AI models grow in power and sophistication, their synergy with experimental biology will be essential for translating computational predictions into tangible insights into gene regulation, disease mechanisms, and therapeutic development.

Conclusion

ML, particularly DL, is beginning to uncover patterns in genetic and epigenetic data that inform disease development. While powerful, DL models must become more interpretable before they can be fully

embraced in clinical decision-making. Their ability to integrate diverse data types, ranging from molecular profiles to electronic health records offers a pathway to truly personalised medicine. AI in epigenetics promises to revolutionise our understanding of genome regulation through advanced models such as DL and transformers that can predict and interpret epigenetic patterns. To fully realise the potential, challenges like data standardisation, model interpretability, and integration of temporal and spatial information must be addressed. Emphasis on explainable AI, cross-species learning, and single-cell or long-read data will further enhance insights, enabling AI and experimental biology to work together toward breakthroughs in gene regulation and disease therapeutics.

Acknowledgements

RS gratefully acknowledges the invaluable support of her Co-Investigator, Prof. Gopalan Rajaraman from the Department of Chemistry, Indian Institute of Technology Bombay, Mumbai, India.

Author contributions

The manuscript is solely written by the corresponding author.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding information

The author acknowledges Department of Science & Technology, Government of India for financial support with reference DST/WIDUSHI-B/CS/2024/35 under WIDUSHI Programme to carry out this work.

ORCID

Ruby Srivastava  0000-0002-2367-0176

Data availability statement

Not Applicable.

Declaration of generative AI and AI-assisted technologies

ChatGPT-4.0 was used to grammatically rephrase the manuscript.

References

- [1] Caudai C, Galizia A, Geraci F, et al. AI applications in functional genomics. *Comput Struct Biotechnol J*. 2021;19:5762–5790. doi: [10.1016/j.csbj.2021.10.009](https://doi.org/10.1016/j.csbj.2021.10.009)
- [2] Zhang Q, Wang Y. AI in biology: transforming genomic research with machine learning. *Computational Molecular Biology*. 2024;14(3):106–114. doi: [10.5376/cmb.2024.14.0013](https://doi.org/10.5376/cmb.2024.14.0013)
- [3] Aghziel A, Mahraz MA, Tairi H, et al. Artificial intelligence for comprehensive DNA methylation analysis: overview, challenges, and future directions. *Brief Bioinform*. 2025;26(5):bbaf468. doi: [10.1093/bib/bbaf468](https://doi.org/10.1093/bib/bbaf468)
- [4] Allis C, Jenuwein T. The molecular hallmarks of epigenetic control. *Nat Rev Genet*. 2016;17:487–500. doi: [10.1038/nrg.2016.59](https://doi.org/10.1038/nrg.2016.59)
- [5] Waddington CH. *Organisers & Genes*. Cambridge: Cambridge University Press; 1940
- [6] Cavalli G, Heard E. Advances in epigenetics link genetics to the environment and disease. *Natur*. 2019;571:489–499. doi: [10.1038/s41586-019-1411-0](https://doi.org/10.1038/s41586-019-1411-0)
- [7] Schaefer J, Lehne M, Schepers J, et al. The use of machine learning in rare diseases: a scoping review. *Orphanet J Rare Dis*. 2020;15:145. doi: [10.1186/s13023-020-01424-6](https://doi.org/10.1186/s13023-020-01424-6)
- [8] Wei S, Tao J, Xu J, et al. Ten years of EWAS. *Adv Sci (Weinh)*. 2021;8:e2100727. doi: [10.1002/advs.202100727](https://doi.org/10.1002/advs.202100727)
- [9] Ehrlich M. DNA hypermethylation in disease: mechanisms and clinical relevance. *Epigenetics*. 2019;14(12):1141–1163. doi: [10.1080/15592294.2019.1638701](https://doi.org/10.1080/15592294.2019.1638701)

- [10] Srivastava R. Multiomics analysis on the clinical treatment for multiple myeloma (MM). *J Hematol Mult Myeloma*. 2023;6:1028.
- [11] Srivastava R. Applications of artificial intelligence in medicine. *Explor Res Hypothesis Med*. 2024;9:138–146. doi: [10.14218/ERHM.2023.00048](https://doi.org/10.14218/ERHM.2023.00048)
- [12] Geissler F, Nesic K, Kondrashova O, et al. The role of aberrant DNA methylation in cancer initiation and clinical impacts. *Ther Adv Med Oncol*. 2024;16:17588359231220511. doi: [10.1177/17588359231220511](https://doi.org/10.1177/17588359231220511)
- [13] Jin Z, Liu Y. DNA methylation in human diseases. *Genes Dis*. 2018;5(1):1–8. doi: [10.1016/j.gendis.2018.01.002](https://doi.org/10.1016/j.gendis.2018.01.002)
- [14] Liu R, Zhao E, Yu H, et al. Methylation across the central dogma in health and diseases: new therapeutic strategies. *Sig Transduct Target Ther*. 2023;8:310. doi: [10.1038/s41392-023-01528-y](https://doi.org/10.1038/s41392-023-01528-y)
- [15] Adusumalli S, Omar MFM, Soong R, Benoukraf T. Methodological aspects of whole-genome bisulfite sequencing analysis. *Brief Bioinform*. 2015;16(3):369–379. doi: [10.1093/bib/bbu016](https://doi.org/10.1093/bib/bbu016)
- [16] Ligthart S, Marzi C, Aslibekyan S, et al. DNA methylation signatures of chronic low-grade inflammation are associated with complex diseases. *Genome Biol*. 2016;17:255. doi: [10.1186/s13059-016-1119-5](https://doi.org/10.1186/s13059-016-1119-5)
- [17] Trimarchi MP, Murphy M, Frankhouser D, et al. Enrichment-based DNA methylation analysis using next-generation sequencing: sample exclusion, estimating changes in global methylation, and the contribution of replicate lanes. *BMC Genomics*. 2012;13(Suppl 8):S6. doi: [10.1186/1471-2164-13-S8-S6](https://doi.org/10.1186/1471-2164-13-S8-S6)
- [18] Gong T, Borgard H, Zhang Z, et al. Analysis and performance assessment of the whole genome bisulfite sequencing data workflow: currently available tools and a practical guide to advance DNA methylation studies. *Small Methods*. 2022;6(3):e2101251. doi: [10.1002/smt.202101251](https://doi.org/10.1002/smt.202101251)
- [19] Srivastava R. Role of transcriptomics in precision oncology. *Rep Radiother Oncol*. 2022;9:e142195. doi: [10.5812/rro-142195](https://doi.org/10.5812/rro-142195)
- [20] Srivastava R. Artificial intelligence multiomics in precision oncology. Newcastle upon Tyne: Cambridge Scholars Publishing; 2024
- [21] Srivastava R. Applications of artificial intelligence multiomics in precision oncology. *J Cancer Res Clin Oncol*. 2023;149:503–510. doi: [10.1007/s00432-022-04161-4](https://doi.org/10.1007/s00432-022-04161-4)
- [22] Kurdyukov S, Bullock M. DNA methylation analysis: choosing the right method. *Biology (Basel)*. 2016;5:3. doi: [10.3390/biology5010003](https://doi.org/10.3390/biology5010003)
- [23] Rauschert S, Raubenheimer K, Melton PE, et al. Machine learning and clinical epigenetics: a review of challenges for diagnosis and classification. *Clin Epigenetics*. 2020;12:51. doi: [10.1186/s13148-020-00842-4](https://doi.org/10.1186/s13148-020-00842-4)
- [24] Pidsley R, Zotenko E, Peters TJ, et al. Critical evaluation of the illumina methylationEPIC beadchip microarray for whole-genome DNA methylation profiling. *Genome Biol*. 2016;17:208. doi: [10.1186/s13059-016-1066-1](https://doi.org/10.1186/s13059-016-1066-1)
- [25] Liu J, Siegmund KD. An evaluation of processing methods for humanmethylation450 beadchip data. *BMC Genomics*. 2016;17:469. doi: [10.1186/s12864-016-2819-7](https://doi.org/10.1186/s12864-016-2819-7)
- [26] Carreras-Gallo N, Dwaraka VB, Jima DD, et al. Creation and validation of the first infinium DNA methylation array for the human imprintome. *bioRxiv [Preprint]*. 2024 Jan 16:2024.01.15.575646. doi: [10.1101/2024.01.15.575646](https://doi.org/10.1101/2024.01.15.575646)
- [27] Andrews SV, Ladd-Acosta C, Feinberg AP, et al. “Gap hunting” to characterize clustered probe signals in Illumina methylation array data. *Epigenetics & Chromatin*. 2016;9:56. doi: [10.1186/s13072-016-0107-z](https://doi.org/10.1186/s13072-016-0107-z)
- [28] Price EM, Cotton AM, Lam LL, et al. Additional annotation enhances potential for biologically-relevant analysis of the Illumina Infinium HumanMethylation450 BeadChip array. *Epigenetics & Chromatin*. 2013;6:4. doi: [10.1186/1756-8935-6-4](https://doi.org/10.1186/1756-8935-6-4)
- [29] https://www.cd-genomics.com/resource-methylation-arrays-evolution-basic-profiling-high-resolution.html?utm_source/.
- [30] Payne SR. From discovery to the clinic: the novel DNA methylation biomarker mSEPT9 for the detection of colorectal cancer in blood. *Epigenomics*. 2010;2:575–585. doi: [10.2217/epi.10.35](https://doi.org/10.2217/epi.10.35)
- [31] Kagan AB, Garrison DA, Anders NM, et al. DNA methyltransferase inhibitor exposure-response: challenges and opportunities. *Clin Transl Sci*. 2023;16:1309–1322. doi: [10.1111/cts.13548](https://doi.org/10.1111/cts.13548)
- [32] Srivastava R. Advancing precision oncology with AI-powered genomic analysis. *Front Pharmacol*. 2025;16:1591696. doi: [10.3389/fphar.2025.1591696](https://doi.org/10.3389/fphar.2025.1591696)
- [33] Srivastava R. Chemical reactivity and optical and pharmacokinetics studies of 14 multikinase inhibitors and their docking interactions toward ACK1 for precision oncology. *Front Chem*. 2022;10. doi: [10.3389/fchem.2022.843642](https://doi.org/10.3389/fchem.2022.843642)
- [34] Yassi M, Chatterjee A, Parry M. Application of deep learning in cancer epigenetics through DNA methylation analysis. *Brief Bioinform*. 2023;24:bbad411. doi: [10.1093/bib/bbad411](https://doi.org/10.1093/bib/bbad411)
- [35] Dankar FK, Ptitsyn A, Dankar SK. The development of large-scale de-identified biomedical databases in the age of genomics—principles and challenges. *Hum Genomics*. 2018;12:19. doi: [10.1186/s40246-018-0147-5](https://doi.org/10.1186/s40246-018-0147-5)
- [36] <https://www.genome.gov/about-genomics/fact-sheets/Genomic-Data-Science?>
- [37] Moran S, Arribas C, Esteller M. Validation of a DNA methylation microarray for 850,000 CpG sites of the human genome enriched in enhancer sequences. *Epigenomics*. 2016;8(3):389–399. doi: [10.2217/epi.15.114](https://doi.org/10.2217/epi.15.114)
- [38] <https://ega-archive.org/access/data-access-committee/dac-portal/>
- [39] Sona P, Hong JH, Lee S, et al. Integrated genome sizing (IGS) approach for the parallelization of whole genome analysis. *BMC Bioinf*. 2018;19:462. doi: [10.1186/s12859-018-2499-1](https://doi.org/10.1186/s12859-018-2499-1)
- [40] Elisseev V, Gardiner LJ, Krishna R. Scalable in-memory processing of omics workflows. *Comput Struct Biotechnol J*. 2022;20:1914–1924. doi: [10.1016/j.csbj.2022.04.014](https://doi.org/10.1016/j.csbj.2022.04.014).

- [41] Toh TS, Dondelinger F, Wang D. Looking beyond the hype: applied AI and machine learning in translational medicine. *EBioMedicine*. 2019;47:607–615. doi: [10.1016/j.ebiom.2019.08.027](https://doi.org/10.1016/j.ebiom.2019.08.027)
- [42] Ben-David A. Comparison of classification accuracy using Cohen's Weighted Kappa. *Expert Syst Appl*. 2008;34:825–832. doi: [10.1016/j.eswa.2006.10.022](https://doi.org/10.1016/j.eswa.2006.10.022)
- [43] Sokolova M, Lapalme G. A systematic analysis of performance measures for classification tasks. *Inf Process Manag*. 2009;45:427–437. doi: [10.1016/j.ipm.2009.03.002](https://doi.org/10.1016/j.ipm.2009.03.002)
- [44] Haixiang G, Yijing L, Shang J, et al. Learning from class-imbalanced data: review of methods and applications. *Expert Syst Appl*. 2017;73:220–239. doi: [10.1016/j.eswa.2016.12.035](https://doi.org/10.1016/j.eswa.2016.12.035)
- [45] Aref-Eshghi E, Schenkel LC, Ainsworth P, et al. Genomic DNA methylation-derived algorithm enables accurate detection of malignant prostate tissues. *Front Oncol*. 2018;8:100. doi: [10.3389/fonc.2018.00100](https://doi.org/10.3389/fonc.2018.00100)
- [46] Capper D, Jones DT, Sill M, et al. DNA methylation-based classification of central nervous system tumours. *Nature*. 2018;555:469–474. doi: [10.1038/nature26000](https://doi.org/10.1038/nature26000)
- [47] Dogan MV, Grumbach IM, Michaelson JJ, et al. Integrated genetic and epigenetic prediction of coronary heart disease in the framingham heart study. *PLoS One*. 2018;13:e0190549. doi: [10.1371/journal.pone.0190549](https://doi.org/10.1371/journal.pone.0190549)
- [48] Tarca AL, Carey VJ, Chen X-W, et al. Machine learning and its applications to biology. *PLoS Comput Biol*. 2007;3:e116. doi: [10.1371/journal.pcbi.0030116](https://doi.org/10.1371/journal.pcbi.0030116)
- [49] Boulesteix A-L, Strimmer K. Partial least squares: a versatile tool for the analysis of high-dimensional genomic data. *Brief Bioinform*. 2006;8:32–44. doi: [10.1093/bib/bbl016](https://doi.org/10.1093/bib/bbl016)
- [50] Park H-S, Jun C-H. A simple and fast algorithm for K-medoids clustering. *Expert Syst Appl*. 2009;36:3336–3341. doi: [10.1016/j.eswa.2008.01.039](https://doi.org/10.1016/j.eswa.2008.01.039)
- [51] Brasil S, Neves CJ, Rijoff T, et al. Artificial intelligence in epigenetic studies: shedding light on rare diseases. *Front Mol Biosci*. 2021;8:648012. doi: [10.3389/fmolb.2021.648012](https://doi.org/10.3389/fmolb.2021.648012)
- [52] Karim MR, Beyan O, Zappa A, et al. Deep learning-based clustering approaches for bioinformatics. *Brief Bioinform*. 2021;22(1):393–415. doi: [10.1093/bib/bbz170](https://doi.org/10.1093/bib/bbz170)
- [53] Oyelade J, Isewon I, Oladipupo F, et al. Clustering algorithms: their application to gene expression data. *Bioinform Biol Insights*. 2016;10:237–253. doi: [10.4137/BBI.S38316](https://doi.org/10.4137/BBI.S38316)
- [54] Hormaty S, Seiwan AN, Rasheed BH, et al. A review on biomarker-enhanced machine learning for early diagnosis and outcome prediction in ovarian cancer management. *Cancer Med*. 2025;14(17):e71224. doi: [10.1002/cam4.71224](https://doi.org/10.1002/cam4.71224)
- [55] Trezza A, Visibelli A, Roncaglia B, et al. Unsupervised Learning in precision medicine: unlocking personalized healthcare through AI. *Appl Sci*. 2024;14:9305. doi: [10.3390/app14209305](https://doi.org/10.3390/app14209305)
- [56] Xu J, Wu P, Chen Y, et al. Cancer subtype classification via hierarchical deep learning integration of multi-omics data. *BMC Bioinf*. 2019;20:527. doi: [10.1186/s12859-019-3106-9](https://doi.org/10.1186/s12859-019-3106-9)
- [57] Baldi P. Autoencoders, unsupervised learning, and deep architectures. *Proc ICML Workshop Unsupervised Transfer Learn*. 2012. p. 37–49. <http://proceedings.mlr.press/v27/baldi12a.html>
- [58] The Cancer Genome Atlas (TCGA). National Cancer Institute. <https://www.cancer.gov/tcga>
- [59] ENCODE Project Consortium, An integrated encyclopedia of DNA elements in the human genome. *Nature*. 2012;489:57–74. doi: [10.1038/nature11247](https://doi.org/10.1038/nature11247)
- [60] Fernández JM, de la Torre V, Richardson D, et al. The BLUEPRINT data analysis portal. *Cell Syst*. 2016;3(5):491–495.e5. doi: [10.1016/j.cels.2016.10.021](https://doi.org/10.1016/j.cels.2016.10.021)
- [61] Moradpour M, Abdulah SNA. CRISPR/dCas9 platforms in plants: strategies and applications beyond genome editing. *Plant Biotechnol J*. 2020;18(1):32–44. doi: [10.1111/pbi.13232](https://doi.org/10.1111/pbi.13232)
- [62] Qian J, Liu SX. DNA methylation editing using CRISPR/dCas9-Tet1. *Bio Protoc*. 2024;14:e4976. doi: [10.21769/BioProtoc.4976](https://doi.org/10.21769/BioProtoc.4976)
- [63] Cheng MW, Mitra M, Collier HA. Epigenetic factor expression landscape and tumor prognosis. *Commun Biol*. 2023;6:1138. doi: [10.1038/s42003-023-05459-w](https://doi.org/10.1038/s42003-023-05459-w)
- [64] Rasmussen KD, Helin K. Role of TET enzymes in DNA methylation, development, and cancer. *Genes Dev*. 2016;30(7):733–750. doi: [10.1101/gad.276568.115](https://doi.org/10.1101/gad.276568.115)
- [65] Boratyn GM, Thierry-Mieg J, Thierry-Mieg D, et al. Magic-BLAST, an accurate RNA-seq aligner for long and short reads. *BMC Bioinf*. 2019;20:405. doi: [10.1186/s12859-019-2996-x](https://doi.org/10.1186/s12859-019-2996-x)
- [66] Orabi B, Erhan E, McConeghy B, et al. Alignment-free clustering of UMI tagged DNA molecules. *Bioinformatics*. 2019;35(11):1829–1836. doi: [10.1093/bioinformatics/bty888](https://doi.org/10.1093/bioinformatics/bty888)
- [67] Li Heng, Durbin Richard. Fast and accurate short read alignment with Burrows–wheeler transform. *Bioinformatics*. 2009;25(14):1754–1760. doi: [10.1093/bioinformatics/btp324](https://doi.org/10.1093/bioinformatics/btp324)
- [68] Langmead B, Salzberg S. Fast gapped-read alignment with Bowtie 2. *Nat Methods*. 2012;9:357–359. doi: [10.1038/nmeth.1923](https://doi.org/10.1038/nmeth.1923)
- [69] Kim D, Paggi JM, Park C, et al. Graph-based genome alignment and genotyping with HISAT2 and HISAT-genotype. *Nat Biotechnol*. 2019;37:907–915. doi: [10.1038/s41587-019-0201-4](https://doi.org/10.1038/s41587-019-0201-4)
- [70] Krueger Felix, Andrews Simon R. Bismark: a flexible aligner and methylation caller for Bisulfite-Seq applications. *Bioinformatics*. 2011;27(11):1571–1572. doi: [10.1093/bioinformatics/btr167](https://doi.org/10.1093/bioinformatics/btr167)
- [71] Brasil S, Neves CJ, Rijoff T, et al. Artificial intelligence in epigenetic studies: shedding light on rare diseases. *Front Mol Biosci*. 2021;8:648012. doi: [10.3389/fmolb.2021.648012](https://doi.org/10.3389/fmolb.2021.648012)

- [72] Roman-Naranjo P, Parra-Perez AM, Lopez-Escamez JA. A systematic review on machine learning approaches in the diagnosis and prognosis of rare genetic diseases. *J Biomed Inform.* **2023**;143:104429. doi: [10.1016/j.jbi.2023.104429](https://doi.org/10.1016/j.jbi.2023.104429)
- [73] Bourdon V, Philippe C, Labrune O, et al. A detailed analysis of the MECP2 gene: prevalence of recurrent mutations and gross DNA rearrangements in Rett syndrome patients. *Qual Life Res.* **2001**;108:43–50. doi: [10.1007/s004390000422](https://doi.org/10.1007/s004390000422)
- [74] Pablo MJ, Pamplona P, Haddad M, et al. High rate of autonomic neuropathy in Cornelia de Lange Syndrome. *Orphanet J Rare Dis.* **2021**;16(1):458. doi: [10.1186/s13023-021-02082-y](https://doi.org/10.1186/s13023-021-02082-y)
- [75] Mendiola AJP, LaSalle JM. Epigenetics in Prader-Willi Syndrome. *Front Genet.* **2021**;12:624581. doi: [10.3389/fgene.2021.624581](https://doi.org/10.3389/fgene.2021.624581)
- [76] Cecere F, Pignata L, D'Angelo E, et al. Highly variable genomic methylation in the Beckwith-Wiedemann syndrome associated with multi-locus imprinting disturbances. *Clin Epigenet.* **2025**;17:160. doi: [10.1186/s13148-025-01971-4](https://doi.org/10.1186/s13148-025-01971-4)
- [77] Kovaka S, Hook PW, Jenike KM, et al. Uncalled4 improves nanopore DNA and RNA modification detection via fast and accurate signal alignment. *Nat Methods.* **2025**;22:681–691. doi: [10.1038/s41592-025-02631-4](https://doi.org/10.1038/s41592-025-02631-4)
- [78] Flusberg BA, Webster DR, Lee JH, et al. Direct detection of DNA methylation during single-molecule, real-time sequencing. *Nat Methods.* **2010**;7(6):461–465. doi: [10.1038/nmeth.1459](https://doi.org/10.1038/nmeth.1459)
- [79] Moradpour M, Abdullah SNA. CRISPR/dCas9 platforms in plants: strategies and applications beyond genome editing. *Plant Biotechnol J.* **2020**;18(1):32–44. doi: [10.1111/pbi.13232](https://doi.org/10.1111/pbi.13232)
- [80] Qian J, Liu SX. DNA methylation editing using CRISPR/dCas9-Tet1. *Bio Protoc.* **2024**;14:e4976. doi: [10.21769/BioProtoc.4976](https://doi.org/10.21769/BioProtoc.4976)
- [81] Cheng MW, Mitra M, Collier HA. Epigenetic factor expression landscape and tumor prognosis. *Commun Biol.* **2023**;6:1138. doi: [10.1038/s42003-023-05459-w](https://doi.org/10.1038/s42003-023-05459-w)
- [82] Taye MM. Theoretical understanding of convolutional neural network: concepts, architectures, applications, future directions. *Computation.* **2023**;11(3):52. doi: [10.3390/computation11030052](https://doi.org/10.3390/computation11030052)
- [83] Baldi P. Autoencoders, unsupervised learning, and deep architectures. *Proc ICML Workshop Unsupervised Transfer Learn.* 27. PMLR. **2012**. p. 37–49. <http://proceedings.mlr.press/v27/baldi12a.html>
- [84] Liu P, Qiu X, Huang X. Recurrent neural network for text classification with multi-task learning. *arXiv preprint.* **2016**. arXiv:1605.05101.
- [85] Hochreiter S, Schmidhuber J. Long short-term memory. *Neural Comput.* **1997**;9:1735–1780. doi: [10.1162/neco.1997.9.8.1735](https://doi.org/10.1162/neco.1997.9.8.1735)
- [86] Ma J, Huang X, Wu H, et al. Bidirectional gated recurrent unit (BiGRU)-based model for concrete gravity dam displacement prediction. *Sustainability.* **2025**;17:7401. doi: [10.3390/su17167401](https://doi.org/10.3390/su17167401)
- [87] Choi SR, Lee M. Transformer architecture and attention mechanisms in genome data analysis: a comprehensive review. *Biology (Basel).* **2023**;12(7):1033. doi: [10.3390/biology12071033](https://doi.org/10.3390/biology12071033)
- [88] Albaradei S, Napolitano F, Thafar MA, et al. MetaCancer: a deep learning-based pan-cancer metastasis prediction model developed using multi-omics data. *Comput Struct Biotechnol J.* **2021**;19:4404–4411. doi: [10.1016/j.csbj.2021.08.006](https://doi.org/10.1016/j.csbj.2021.08.006)
- [89] Gomes R, Paul N, He N, et al. Application of feature selection and deep learning for cancer prediction using DNA methylation markers. *Genes (Basel).* **2022**;13(9):1557. doi: [10.3390/genes13091557](https://doi.org/10.3390/genes13091557)
- [90] Jabari S, Kobow K, Pieper T, et al. DNA methylation-based classification of malformations of cortical development in the human brain. *Acta Neuropathol.* **2022**;143:93–104. doi: [10.1007/s00401-021-02386-0](https://doi.org/10.1007/s00401-021-02386-0)
- [91] Orozco JIJ, Knijnenburg TA, Manughian-Peter AO, et al. Epigenetic profiling for the molecular classification of metastatic brain tumors. *Nat Commun.* **2018**;9(1):4627. doi: [10.1038/s41467-018-06715-y](https://doi.org/10.1038/s41467-018-06715-y)
- [92] Levy JJ, Chen Y, Azizgolshani N, et al. MethylSPWNet and MethylCapsNet: biologically motivated organization of DNAm neural networks, inspired by capsule networks. *NPJ Syst Biol Appl.* **2021** ;7(1):33. doi: [10.1038/s41540-021-00193-7](https://doi.org/10.1038/s41540-021-00193-7)
- [93] Levy JJ, Titus AJ, Petersen CL, et al. MethylNet: an automated and modular deep learning approach for DNA methylation analysis. *BMC Bioinf.* **2020**;21:108. doi: [10.1186/s12859-020-3443-8](https://doi.org/10.1186/s12859-020-3443-8)
- [94] Li J, Wei L, Zhang X, et al. DISMIR: deep learning-based non-invasive cancer detection by integrating DNA sequence and methylation information of individual cell-free DNA reads. *Brief Bioinform.* **2021**;22:1–11. doi: [10.1093/bib/bbab250](https://doi.org/10.1093/bib/bbab250)
- [95] Lin W, Hu S, Wu Z, et al. iCancer-Pred: a tool for identifying cancer and its type using DNA methylation. *Genomics.* **2022**;114:110486. doi: [10.1016/j.ygeno.2022.110486](https://doi.org/10.1016/j.ygeno.2022.110486)
- [96] Liu L, Zeng P, Yang S, et al. Leveraging methylation to identify the potential causal genes associated with survival in lung adenocarcinoma and lung squamous cell carcinoma. *Oncol Lett.* **2020**;20(1):193–200. doi: [10.3892/ol.2020.11564](https://doi.org/10.3892/ol.2020.11564)
- [97] Minh D, Wang HX, Li YF, et al. Explainable artificial intelligence: a comprehensive review. *Artif Intell Rev.* **2022**;55:3503–3568. doi: [10.1007/s10462-021-10088-y](https://doi.org/10.1007/s10462-021-10088-y)
- [98] Sun M, Xu B, Chen C, et al. Tissue of origin prediction for cancer of unknown primary using a targeted methylation sequencing panel. *Clin Epigenet.* **2024**;16:25. doi: [10.1186/s13148-024-01638-6](https://doi.org/10.1186/s13148-024-01638-6)

- [99] Qiu YL, Zheng H, Gevaert O. Genomic data imputation with variational auto-encoders. *Gigascience*. 2020;9(8):giaa082. doi: [10.1093/gigascience/giaa082](https://doi.org/10.1093/gigascience/giaa082)
- [100] Angermueller C, Lee HJ, Reik W, et al. DeepCpG: accurate prediction of single-cell DNA methylation states using deep learning. *Genome Biol*. 2017;18:67. doi: [10.1186/s13059-017-1189-z](https://doi.org/10.1186/s13059-017-1189-z)
- [101] Cheng X, Wang J, Li Q, et al. BiLSTM-5mC: a bidirectional long short-term memory-based approach for predicting 5-methylcytosine sites in genome-wide DNA promoters. *Molecules*. 2021;26:7414. doi: [10.3390/molecules26247414](https://doi.org/10.3390/molecules26247414)
- [102] De Waele G, Clauwaert J, Menschaert G, et al. CpG Transformer for imputation of single-cell methylomes. *Bioinformatics*. 2022;38:597–603. doi: [10.1093/bioinformatics/btab746](https://doi.org/10.1093/bioinformatics/btab746)
- [103] Pan X, Liu B, Wen X, et al. D-GPM: a deep learning method for gene promoter methylation inference. *Genes (Basel)*. 2019;10:807. doi: [10.3390/genes10100807](https://doi.org/10.3390/genes10100807)
- [104] Tian Q, Zou J, Tang J, et al. MRCNN: a deep learning model for regression of genome-wide DNA methylation. *BMC Genomics*. 2019;20:192. doi: [10.1186/s12864-019-5488-5](https://doi.org/10.1186/s12864-019-5488-5)
- [105] Zhou J, Weinberger DR, Han S. Deep learning predicts DNA methylation regulatory variants in specific brain cell types and enhances fine mapping for brain disorders. *Sci Adv*. 2025;11(1):eadn1870. doi: [10.1126/sciadv.adn1870](https://doi.org/10.1126/sciadv.adn1870)
- [106] Bichindaritz I, Liu G, Bartlett C. Integrative survival analysis of breast cancer with gene expression and DNA methylation data. *Bioinformatics*. 2021;37:2601–2608. doi: [10.1093/bioinformatics/btab140](https://doi.org/10.1093/bioinformatics/btab140)
- [107] Gu T, Zhao X. Integrating multi-platform genomic datasets for kidney renal clear cell carcinoma subtyping using stacked denoising autoencoders. *Sci Rep*. 2019;9:16668. doi: [10.1038/s41598-019-53048-x](https://doi.org/10.1038/s41598-019-53048-x)
- [108] Chai H, Zhou X, Zhang Z, et al. Integrating multi-omics data through deep learning for accurate cancer prognosis prediction. *Comput Biol Med*. 2021;134:104481. doi: [10.1016/j.compbiomed.2021.104481](https://doi.org/10.1016/j.compbiomed.2021.104481)
- [109] Ma T, Zhang A. Integrate multi-omics data with biological interaction networks using Multi-view Factorization AutoEncoder (MAE). *BMC Genomics*. 2019;20(Suppl 11):944. doi: [10.1186/s12864-019-6285-x](https://doi.org/10.1186/s12864-019-6285-x)
- [110] Pan X, Burgman B, Wu E, et al. I-modern: an interpretable deep learning framework for identifying therapeutic targets in glioma through multi-omics network modeling. *Comput Struct Biotechnol J*. 2022;20:3511–3521. doi: [10.1016/j.csbj.2022.06.058](https://doi.org/10.1016/j.csbj.2022.06.058)
- [111] Song H, Ruan C, Xu Y, et al. Survival stratification for colorectal cancer via multi-omics integration using an autoencoder-based model. *Exp Biol Med (Maywood)*. 2022;247:898–909. doi: [10.1177/15353702211065010](https://doi.org/10.1177/15353702211065010)
- [112] Takahashi S, Asada K, Takasawa K, et al. Predicting Deep Learning Based Multi-Omics Parallel Integration Survival Subtypes in Lung Cancer Using Reverse Phase Protein Array Data. *Biomolecules*. 2020;10(10):1460. doi: [10.3390/biom10101460](https://doi.org/10.3390/biom10101460)
- [113] Herrmann M, Probst P, Hornung R, et al. Large-scale benchmark study of survival prediction methods using multi-omics data. *Brief Bioinform*. 2021;22(3):bbaa167. doi: [10.1093/bib/bbaa167](https://doi.org/10.1093/bib/bbaa167)
- [114] Peng C, Zheng Y, Huang DS. Breast cancer gene discovery using capsule networks on multi-omics data. *IEEE/ACM Trans Comput Biol Bioinform*. 2020;17:1605–1612. doi: [10.1109/TCBB.2019.2914776](https://doi.org/10.1109/TCBB.2019.2914776)
- [115] Massi MC, Dominoni L, Ieva F, et al. A deep survival EWAS approach estimating risk profile based on pre-diagnostic DNA methylation: an application to breast cancer time to diagnosis. *PLoS Comput Biol*. 2022;18(9):e1009959. doi: [10.1371/journal.pcbi.1009959](https://doi.org/10.1371/journal.pcbi.1009959)
- [116] Choi B-H, Chae K-J. Capsule network based modeling of multi-omics data for discovery of breast cancer-related genes (CapsNetMMD) In: *Proceedings of the IEEE/EMBS International Conference on Biomedical and Health Informatics (BHI)*. 2019. doi: [10.1109/BHI.2019.8834630](https://doi.org/10.1109/BHI.2019.8834630)
- [117] Karim A, Rafi U, Raza K. Rise of deep learning clinical applications and challenges in omics data: a systematic review. *Diagnostics*. 2023;13(4):664. doi: [10.3390/diagnostics13040664](https://doi.org/10.3390/diagnostics13040664)
- [118] Seal DB, Das V, Goswami S, et al. Estimating gene expression from DNA methylation and copy number variation: a deep learning regression model for multi-omics integration. *Genomics*. 2020;112(3):2833–2841. doi: [10.1016/j.ygeno.2020.03.021](https://doi.org/10.1016/j.ygeno.2020.03.021)
- [119] Xu J, Wu P, Chen Y, et al. A hierarchical integration deep flexible neural forest framework for cancer subtype classification by integrating multi-omics data. *BMC Bioinf*. 2019;20:527. doi: [10.1186/s12859-019-3116-7](https://doi.org/10.1186/s12859-019-3116-7)
- [120] de Lima Camillo LP, Lapierre LR, Singh R. A pan-tissue DNA-methylation epigenetic clock based on deep learning. *npj Aging*. 2022;8:4. doi: [10.1038/s41514-022-00085-y](https://doi.org/10.1038/s41514-022-00085-y)
- [121] Lim J, Park C, Kim M, et al. Advances in single-cell omics and multiomics for high-resolution molecular profiling. *Exp Mol Med*. 2024;56(3):515–526. doi: [10.1038/s12276-024-01186-2](https://doi.org/10.1038/s12276-024-01186-2)
- [122] Baek S, Lee I. Single-cell ATAC sequencing analysis: from data preprocessing to hypothesis generation. *Comput Struct Biotechnol J*. 2020;18:1429–1439. doi: [10.1016/j.csbj.2020.06.012](https://doi.org/10.1016/j.csbj.2020.06.012)
- [123] Zhang H, Mulqueen RM, Iannuzo N, et al. txci-ATAC-seq: a massive-scale single-cell technique to profile chromatin accessibility. *Genome Biol*. 2024;25(1):78. doi: [10.1186/s13059-023-03150-1](https://doi.org/10.1186/s13059-023-03150-1)
- [124] Zeng Y, Luo M, Shangquan N, et al. Deciphering cell types by integrating scATAC-seq data with genome sequences. *Nat Comput Sci*. 2024;4:285–298. doi: [10.1038/s43588-024-00622-7](https://doi.org/10.1038/s43588-024-00622-7)
- [125] Ahn J, Heo S, Lee J, et al. Introduction to single-cell dna methylation profiling methods. *Biomolecules*. 2021;11(7):1013. doi: [10.3390/biom11071013](https://doi.org/10.3390/biom11071013)

- [126] Sherif ZA, Ogunwobi OO, Ressom HW. Mechanisms and technologies in cancer epigenetics. *Front Oncol.* **2025**;14:1513654. doi: [10.3389/fonc.2024.1513654](https://doi.org/10.3389/fonc.2024.1513654)
- [127] Argelaguet R, Arnol D, Bredikhin D, et al. MOFA+: a framework for integration of multimodal single-cell data. *Genome Biol.* **2020**;21:111. doi: [10.1186/s13059-020-02015-1](https://doi.org/10.1186/s13059-020-02015-1)
- [128] Tripathi A, Waqas A, Venkatesan K, et al. Building flexible, scalable, and machine learning-ready multimodal oncology datasets. *Senso.* **2024**;24:1634. doi: [10.3390/s24051634](https://doi.org/10.3390/s24051634)
- [129] Slobodyanyuk M, Bahcheli AT, Klein ZP, et al. Directional integration and pathway enrichment analysis for multi-omics data. *Nat Commun.* **2024**;15:5690. doi: [10.1038/s41467-024-49986-4](https://doi.org/10.1038/s41467-024-49986-4)
- [130] Lee JY, Park W, Kim H, et al. Multi-omics analysis sandbox toolkit for swift derivations of clinically relevant genesets and biomarkers. *BMB Rep.* **2024 Dec**;57(12):521–526. doi: [10.5483/BMBRep.2023-0155](https://doi.org/10.5483/BMBRep.2023-0155)
- [131] Głowacka-Walas J, Sijko K, Wojdan K, et al. playOmics: a multi-omics pipeline for interpretable predictions and biomarker discovery. *BioRxiv.* **2024**. doi: [10.1101/2024.03.12.584088](https://doi.org/10.1101/2024.03.12.584088)
- [132] Settelmeier J, Goetze S, Boshart J, et al. MultiOmicsAgent: guided extreme gradient-boosted decision trees-based approaches for biomarker-candidate discovery in multi-omics data. *bioRxiv.* **2024**. doi: [10.1101/2024.07.24.604727](https://doi.org/10.1101/2024.07.24.604727)
- [133] Clark SJ, Argelaguet R, Kapourani CA, et al. scNMT-seq enables joint profiling of chromatin accessibility DNA methylation and transcription in single cells. *Nat Commun.* **2018**;9:781. doi: [10.1038/s41467-018-03149-4](https://doi.org/10.1038/s41467-018-03149-4)
- [134] Kupershmidt Y, Kasif S, Sharan R. SPIDER: constructing cell-type-specific protein-protein interaction networks. *Bioinform Adv.* **2024**;4(1):vbae130. doi: [10.1093/bioadv/vbae130](https://doi.org/10.1093/bioadv/vbae130)
- [135] Lau JW, Lehnert E, Sethi A, et al. Seven bridges CGC team. the cancer genomics cloud: collaborative, reproducible, and democratized-a new paradigm in large-scale computational research. *Cancer Res.* **2017** ;77(21):e3–e6. doi: [10.1158/0008-5472.CAN-17-0387](https://doi.org/10.1158/0008-5472.CAN-17-0387); Erratum in: *Cancer Research.* **2018**;78(17):5179. doi: [10.1158/0008-5472.can-18-2029](https://doi.org/10.1158/0008-5472.can-18-2029)
- [136] Guo H, Zhu P, Guo F, et al. Profiling DNA methylome landscapes of mammalian cells with single-cell reduced-representation bisulfite sequencing. *Nat Protoc.* **2015**;10:645–659. doi: [10.1038/nprot.2015.039](https://doi.org/10.1038/nprot.2015.039)
- [137] Chen H, Su Z, Li R, et al. Single-cell DNA methylome analysis of circulating tumor cells. *Chin J Cancer Res.* **2021**;33(3):391–404. doi: [10.21147/j.issn.1000-9604.2021.03.10](https://doi.org/10.21147/j.issn.1000-9604.2021.03.10)
- [138] Veneziano M, Savini I, Cortellesi E, et al. Bioinformatics strategies in breast cancer research. *Biomolecules.* **2025**;15(10):1409. doi: [10.3390/biom15101409](https://doi.org/10.3390/biom15101409)
- [139] Grosselin K, Durand A, Marsolier J, et al. Single-cell ChIP-seq reveals chromatin state heterogeneity in breast cancer. *Nat Genet.* **2019**;51:1060–1066. doi: [10.1038/s41588-019-0424-9](https://doi.org/10.1038/s41588-019-0424-9)
- [140] <https://epigenie.com/epigenetic-tools-and-databases/>.
- [141] Berglund A, Putney RM, Hamaidi I, et al. Visualization of epigenetic immune pathway dysregulation in cancer. *Exp Mol Med.* **2021**;53:761–771. doi: [10.1038/s12276-021-00612-z](https://doi.org/10.1038/s12276-021-00612-z)
- [142] Aref-Eshghi E, Abadi AB, Farhadieh ME, et al. DNA methylation and machine learning: challenges and perspective toward enhanced clinical diagnostics. *Clin Epigenetics.* **2025**;17(1):170. doi: [10.1186/s13148-025-01967-0](https://doi.org/10.1186/s13148-025-01967-0)
- [143] Sepulveda AR, Jones D, Ogino S, et al. CpG methylation analysis--current status of clinical assays and potential applications in molecular diagnostics: a report of the association for molecular pathology. *J Mol Diagn.* **2009**;11(4):266–278. doi: [10.2353/jmoldx.2009.080125](https://doi.org/10.2353/jmoldx.2009.080125)
- [144] Chenarani N, Emamjomeh A, Allahverdi A, et al. Survey of bioinformatics tools for epigenetic data analysis. *Genomics.* **2021**;113:1098–1113. doi: [10.1016/j.ygeno.2021.03.004](https://doi.org/10.1016/j.ygeno.2021.03.004)
- [145] Ding W, Chen J, Feng G, et al. DNMIIVD: interactive database for DNA methylation. *Nucleic Acids Res.* **2020**;48:D856–D862. doi: [10.1093/nar/gkz1034](https://doi.org/10.1093/nar/gkz1034)
- [146] Li Y, Ge D, Lu C. SMART App for interactive methylation analysis. *Epigenet Chromatin.* **2019**;12:71. doi: [10.1186/s13072-019-0316-3](https://doi.org/10.1186/s13072-019-0316-3)
- [147] Díez-Villanueva A, Mallona I, Peinado MA. Wanderer: methylation and gene expression viewer in cancer. *Epigenet Chromatin.* **2015**;8:22. doi: [10.1186/s13072-015-0014-6](https://doi.org/10.1186/s13072-015-0014-6)
- [148] Koch A, De Meyer T, Jeschke J, et al. MEXPRESS: visualization of TCGA expression and methylation data. *BMC Genomics.* **2015**;16:636. doi: [10.1186/s12864-015-1847-z](https://doi.org/10.1186/s12864-015-1847-z)
- [149] Modhukur V, Iljasenko T, Metsalu T, et al. MethSurv: a web tool to perform multivariable survival analysis using DNA methylation data. *Epigenomics.* **2018**;10(3):277–288. doi: [10.2217/epi-2017-0118](https://doi.org/10.2217/epi-2017-0118)
- [150] Grunau C, Renault E, Rosenthal A, et al. MethDB--a public database for DNA methylation data. *Nucleic Acids Res.* **2001**;29(1):270–274. doi: [10.1093/nar/29.1.270](https://doi.org/10.1093/nar/29.1.270)
- [151] Li LC, Dahiya R. MethPrimer: designing primers for methylation PCRs. *Bioinformatics.* **2002 Nov**;18(11):1427–1431. doi: [10.1093/bioinformatics/18.11.1427](https://doi.org/10.1093/bioinformatics/18.11.1427)
- [152] Grunau C, Schattevoy R, Mache N, et al. MethTools--a toolbox to visualize and analyze DNA methylation data. *Nucleic Acids Res.* **2000**;28(5):1053–1058. doi: [10.1093/nar/28.5.1053](https://doi.org/10.1093/nar/28.5.1053)
- [153] Fang YC, Huang HC, Juan HF. MeInfoText: associated gene methylation and cancer information from text mining. *BMC Bioinf.* **2008**;9:22. doi: [10.1186/1471-2105-9-22](https://doi.org/10.1186/1471-2105-9-22)
- [154] He X, Chang S, Zhang J, et al. MethyCancer: the database of human DNA methylation and cancer. *Nucleic Acids Res.* **2007**;36(Database issue):D836–41. doi: [10.1093/nar/gkm730](https://doi.org/10.1093/nar/gkm730). 2008 Jan.

- [155] Lv Jie, Liu Hongbo, Su Jianzhong, et al. DiseaseMeth: a human disease methylation database. *Nucleic Acids Res.* **2012**;40(D1):D1030–D1035. doi: [10.1093/nar/gkr1169](https://doi.org/10.1093/nar/gkr1169). 1.
- [156] Xin Y, Chanrion B, O'Donnell AH, et al. MethylomeDB: a database of DNA methylation profiles of the brain. *Nucleic Acids Res.* **2012**;40(Database issue):D1245–9. doi: [10.1093/nar/gkr1193](https://doi.org/10.1093/nar/gkr1193)
- [157] Song Q, Decato B, Hong E, et al. A reference methylome database and analysis pipeline to facilitate integrative and comparative epigenomics. *PLoS One.* **2013**;8(12):e81148. doi: [10.1371/journal.pone.0081148](https://doi.org/10.1371/journal.pone.0081148)
- [158] Geisen S, Barturen G, Alganza ÁM, et al. NGSmethDB: an updated genome resource for high quality, single-cytosine resolution methylomes. *Nucleic Acids Res.* **2014**;42(Database issue):D53–9. doi: [10.1093/nar/gkt1202](https://doi.org/10.1093/nar/gkt1202)
- [159] Liu H, Zhu R, Lv J, et al. DevMouse, the mouse developmental methylome database and analysis tools. *Database (Oxford).* **2014**;2014:bat084–bat084. doi: [10.1093/database/bat084](https://doi.org/10.1093/database/bat084)
- [160] Nagpal G, Sharma M, Kumar S, et al. PCMDB: pancreatic cancer methylation database. *Sci Rep.* **2014**;4:4197. doi: [10.1038/srep04197](https://doi.org/10.1038/srep04197)
- [161] Bin Raies A, Mansour H, Incitti R, et al. DDMGD: the database of text-mined associations between genes methylated in diseases from different species. *Nucleic Acids Res.* **2015**;43(Database issue):D879–86. doi: [10.1093/nar/gku1168](https://doi.org/10.1093/nar/gku1168) Epub 2014 Nov 14.
- [162] Huang HY, Li J, Tang Y, et al. MethHC 2.0: information repository of DNA methylation and gene expression in human cancer. *Nucleic Acids Res.* **2021**;49(D1):D1268–D1275. doi: [10.1093/nar/gkaa1104](https://doi.org/10.1093/nar/gkaa1104)
- [163] Zhang M, Zong W, Zou D, et al. MethBank 4.0: an updated database of DNA methylation across a variety of species. *Nucleic Acids Res.* **2023**;51(D1):D208–D216. doi: [10.1093/nar/gkac969](https://doi.org/10.1093/nar/gkac969)
- [164] Li L, An Y, Ma L, et al. Msuite2: All-in-one DNA methylation data analysis toolkit with enhanced usability and performance. *Comput Struct Biotechnol J.* **2022**;20:1271–1276. doi: [10.1016/j.csbj.2022.03.005](https://doi.org/10.1016/j.csbj.2022.03.005)
- [165] Salam Wassim, Wojewodziec Marcin W, Frisch Dagmar. FAIRification of the DMRichR pipeline: advancing epigenetic research on environmental and evolutionary model organisms. *Bioinformatics Advances.* **2025**;5(1):vbaf024. doi: [10.1093/bioadv/vbaf024](https://doi.org/10.1093/bioadv/vbaf024)
- [166] <https://www.benlauffer.com/CpGMe/>.
- [167] Sullivan S, Sink DW, Trout KL, et al. The Histone Database. *Nucleic Acids Res.* **2002**;30(1):341–342. doi: [10.1093/nar/30.1.341](https://doi.org/10.1093/nar/30.1.341)
- [168] van Steensel B. Chromatin: constructing the big picture. *EMBO J.* **2011**;30(10):1885–1895. doi: [10.1038/emboj.2011.135](https://doi.org/10.1038/emboj.2011.135)
- [169] Zhang Y, Lv J, Liu H, et al. HHMD: the human histone modification database. *Nucleic Acids Res.* **2010**;38(Database issue):D149–54. doi: [10.1093/nar/gkp968](https://doi.org/10.1093/nar/gkp968)
- [170] Ongenaert M, Van Neste L, De Meyer T, et al. PubMeth: a cancer methylation database combining text-mining and expert annotation. *Nucleic Acids Res.* **2007**;36(Database issue):D842–6. doi: [10.1093/nar/gkm788](https://doi.org/10.1093/nar/gkm788). 2008 Jan.
- [171] Choi J, Kim KT, Huh A, et al. dbHiMo: a web-based epigenomics platform for histone-modifying enzymes. *Database (Oxford).* **2015**;2015:bav052–bav052. doi: [10.1093/database/bav052](https://doi.org/10.1093/database/bav052)
- [172] Manu DM, Mwinyi J, Schiöth HB. Challenges in analyzing functional epigenetic data in perspective of adolescent psychiatric health. *Int J Mol Sci.* **2022**;23(10):5856. doi: [10.3390/ijms23105856](https://doi.org/10.3390/ijms23105856)
- [173] Kahn A, Fraga MF. Epigenetics and aging: status, challenges, and needs for the future. *J Gerontol A Biol Sci Med Sci.* **2009**;64(2):195–198. doi: [10.1093/gerona/gln064](https://doi.org/10.1093/gerona/gln064)
- [174] Dai W, Qiao X, Fang Y, et al. Epigenetics-targeted drugs: current paradigms and future challenges. *Sig Transduct Target Ther.* **2024**;9:332. doi: [10.1038/s41392-024-02039-0](https://doi.org/10.1038/s41392-024-02039-0)
- [175] Ren L, Yang Y, Li W, et al. Recent advances in epigenetic anticancer therapeutics and future perspectives. *Front Genet.* **2023**;13:1085391. doi: [10.3389/fgene.2022.1085391](https://doi.org/10.3389/fgene.2022.1085391)
- [176] Zhang C, Xu J, Tang R, et al. Novel research and future prospects of artificial intelligence in cancer diagnosis and treatment. *J Hematol Oncol.* **2023**;16(1):114. doi: [10.1186/s13045-023-01514-5](https://doi.org/10.1186/s13045-023-01514-5)
- [177] Sahoo K, Lingasamy P, Khatun M, et al. Artificial Intelligence in cancer epigenomics: a review on advances in pan-cancer detection and precision medicine. *Epigenetics & Chromatin.* **2025**;18:35. doi: [10.1186/s13072-025-00595-5](https://doi.org/10.1186/s13072-025-00595-5)
- [178] De Waele G, Clauwaert J, Menschaert G, et al. CpG Transformer for imputation of single-cell methylomes. *Bioinformatics.* **2022**;38:597–603. doi: [10.1093/bioinformatics/btab746](https://doi.org/10.1093/bioinformatics/btab746)